

Technical Verification Companion to the ECH Spin-Torsion Program: Λ CDM+ ΔN_{eff} MCMC Proxy, NaMaster Pipeline Recovery, and a Birefringence Consistency Check with a Spectator-ALP Model

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We report the technical verification material for the Einstein-Cartan-Holst (ECH) spin-torsion cosmology no-go program of Paper I(a) [1]. Three analyses are documented. (1) *Stock-CAMB Λ CDM+ ΔN_{eff} MCMC proxy* (Cobaya v3.6.1, **309,189** frozen samples across two converged dataset combinations): this run uses stock CAMB with ΔN_{eff} as a free parameter and carries *no torsion modifications to the Boltzmann equations*; it is reported as a null-consistency test of an extra radiation-like degree of freedom, not as evidence for or against the ECH spin-torsion framework. Both frozen dataset combinations find ΔN_{eff} consistent with zero (-0.020 ± 0.169 full-tension; $+0.058 \pm 0.179$ Planck+BAO+SN) and H_0 consistent with the standard Planck- Λ CDM value (67.68 ± 1.06 full-tension; 67.78 ± 1.09 Planck+BAO+SN, both in $\text{km s}^{-1} \text{Mpc}^{-1}$; the ΔN_{eff} extension does *not* reduce the residual $\sim 3.6\sigma$ tension with the SH0ES local-distance-ladder $H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{Mpc}^{-1}$). (2) *NaMaster pseudo- C_ℓ pipeline validation* on synthetic Λ CDM CMB polarization skies with an ACT-like footprint mask ($N_{\text{side}} = 512$, $\ell_{\text{max}} = 1024$, $f_{\text{sky}} = 0.32$, $10 \mu\text{K} \cdot \text{arcmin}$ white noise, 500 Monte Carlo realizations): injecting the spectator-ALP fiducial value $\beta = 0.27^\circ$ recovers $\hat{\beta} = 0.238^\circ$ (pipeline-recovery bias $\hat{\beta} - \beta_{\text{inj}} = -0.032^\circ$; the worst-case bias across injections, -0.040° at $\beta_{\text{inj}} = 0.342^\circ$, is carried forward as the observed pipeline bias floor — both are MC pipeline-recovery figures, not sky-measurement systematics, and are not directly comparable to each other’s published sky significances). *Scope of the validation*: the test confirms the algebraic pseudo- C_ℓ $E \rightarrow B$ deconvolution under MASTER [2] mode coupling, NOT the physical separation of the cosmic-rotation angle β from the instrumental-miscalibration angle α which strictly requires unrotated galactic foregrounds (the synthetic CMB-only skies contain no galactic foregrounds, so the very component that breaks the β - α degeneracy in published Planck/ACT DR6 measurements is absent by construction). The MC recovery is therefore a pipeline-validation figure, not a sky-detection significance claim. The primary sky detection significance is the published Planck/ACT DR6 2.7 – 2.9σ [3, 4] (the $\beta = 0.342^\circ \pm 0.094^\circ$, 3.6σ headline used throughout this paper is from the published PR3+WMAP9 joint analysis of Eskilt & Komatsu [5]; the PR3-vs-PR4/NPIPE disambiguation is given in fn. 3 in Sec. IV); the pipeline SNR figures refer to recovery of injected MC signals and are *not* competitive sky measurements. (3) *Spectator-ALP consistency check*: for a field with $f_a \sim M_{\text{Pl}}$ (where $M_{\text{Pl}} = (8\pi G)^{-1/2} \approx 2.44 \times 10^{18} \text{ GeV}$ is the *reduced* Planck mass), the scan-prior $m \sim H_0$ region brackets the published joint WMAP+Planck signal $\beta = 0.342^\circ \pm 0.094^\circ$ (3.6σ) [5], but the posterior-supported fixed-coupling ($C_{a\gamma} = 8$) accommodation shifts to $m \gg H_0$ (median $m \simeq 36 H_0$ at fixed $C_{a\gamma} = 8$), and the spectator-safe ($\Omega_a < 0.01$) subset is tuned (see Table IV for the per-subset readout). *Spectator-status caveat*: for $\theta_i \sim 1$ the ALP energy density $\rho_a \sim m^2 f_a^2 \theta_i^2 \sim H_0^2 M_{\text{Pl}}^2$ is of order the critical density today, so the spectator label is only consistent under the explicit restriction $\theta_i \ll 1$ (which constitutes fine-tuning of the misalignment initial condition); at $\theta_i \sim 1$ the ALP must instead be treated as the dark-energy field itself, which is outside the scope of this companion-paper consistency check. The same birefringence arises in standard GR with an identical ALP; it is not a distinctive ECH prediction, and the spectator-status fine-tuning is required regardless of whether the underlying cosmology is a bounce or Λ CDM. All three analyses are compatibility checks only; no analysis here constitutes evidence for or against a bounce cosmology, and all model-preference inference (Bayes factor, $\Delta\text{AIC/BIC}$, nested-sampling evidence) is deferred to follow-up nested-sampling work. A reproducibility manifest is included in Appendix A.

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I. INTRODUCTION

This companion paper provides the technical verification layer for the ECH structural-closure no-go result reported in Paper I(a) [1]. Paper I(a) establishes channel-level closure of four independent minimal-ECH dark-energy routes (kinetic-coupling channel, potential-energy channel, perturbation-transparency channel, and spin-torsion propagator channel) and proves a perturbation-transparency theorem for the Holst sector; the combined structural analysis closes the minimal-ECH parameter space as a dark-energy scenario and motivates the numerical cross-checks of this companion. The present paper documents the three numerical analyses that support and contextualize those results. This is a coordinated concurrent submission: Paper I(a) supplies the physical motivation and the structural no-go result, while the three numerical analyses below — and their data products, chains, and pipeline code — are self-contained and independently reproducible from this paper’s own repository, so no quantitative claim here depends on unpublished companion numerics.

Scope of this paper.—Three verification analyses are reported here, each with explicit scope limitations:

1. Λ CDM+ ΔN_{eff} MCMC proxy (Secs. II and III): stock CAMB Boltzmann solver with ΔN_{eff} as a free parameter. **Not a spin-torsion theory module** (see the §III scope statement); this run tests whether the data prefer an extra radiation-like degree of freedom in standard cosmology.
2. *NaMaster* CMB E-B analysis (Sec. IV): a bias-injection Monte Carlo validation of the pseudo- C_ℓ deconvolution pipeline on synthetic Λ CDM polarization skies. **Not a competitive sky detection.** The high pipeline template-fit SNR figures (e.g., 20.32) refer to

recovery of injected MC signals, not to the significance of the CMB sky measurement, which remains the published 2.7–2.9 σ from Planck/ACT.

3. *Spectator-ALP consistency check* (Sec. VI): a standard GR+ALP computation showing that the observed signal is consistent with an ALP within the scan-prior envelope but near its upper-displacement/coupling edge; the posterior-supported fixed- $C_{a\gamma} = 8$ fit shifts to $m \gg H_0$ (median $\simeq 36 H_0$), well above the prior midpoint. A spectator-safe interpretation requires a tuned misalignment subspace (the $\sim 25\times$ misalignment tuning required to reconcile the observed signal with the spectator-consistent corner is disclosed in Sec. VI and fn. 6). **Not a distinctive ECH prediction.** The same $\beta \approx 0.27^\circ$ arises in any GR+ALP setup with the same parameters; no ECH-specific derivation connects the Holst action to the photon-torsion coupling required.

What is NOT in this paper.—The 13 mechanism-class structural barriers, the perturbation-transparency theorem, the 14-barrier table, and the surviving matter-bounce-specific test predictions ($f_{\text{NL}} = -35/8$, valid strictly for the minimal scalar-only $w = 0$ matter-dominated contraction class, not the broader bouncing-cosmology landscape which encompasses ekpyrotic, Cuscuton, string-gas, and quintom variants with different f_{NL} signatures, and ALP birefringence) are in Paper I(a) [1]. The SPHEREx multi-tracer Fisher forecast (in preparation, [6]; results to be detailed separately) is the subject of Paper II. The multi-survey anomaly catalog (in preparation, [7]) is the subject of Paper III. The galaxy chirality catalog (in preparation, [8]) is the subject of Paper IV.

Cross-paper citations.—When this companion reports MCMC values (H_0 , σ_8 , etc.) that are referenced in the main paper, those values come from Secs. III and V here.

II. COSMOLOGICAL TENSIONS: H_0 AND σ_8

The bounce scenario motivates extending Λ CDM by ΔN_{eff} (particle production at the bounce) as a phenomenological proxy parameter; $(\omega/H)_0$ (angular momentum transfer) and Ω_k are fixed to zero in the actual sampled MCMC configuration of this paper (per the explicit parameter-scope clarification in Sec. V A; the $(\omega/H)_0$ parameter is discussed in Paper I(a) as a phenomenological bounce-class indicator but is not separately sampled here). The full-tension dataset combination includes the SH0ES [9] M_B -anchor likelihood (H0.riess2020Mb, entering through the supernova absolute magnitude rather than as a direct H_0 prior) in the Cobaya likelihood configuration, but the high-precision Planck NPIPE (PR4) CamSpec high- ℓ TT+TEEE + Planck 2018 low- ℓ TT/EE + Planck 2018 lensing likelihoods carry sufficient inverse-variance weight that the posterior H_0 in the proxy run is pulled to

67.68 ± 1.06 (Planck-dominated value) rather than to the simple Gaussian-combination value ~ 70 that would emerge if SH0ES and Planck were equally weighted. We do not therefore claim that the SH0ES tension is resolved or even moved by adding ΔN_{eff} in stock CAMB; in this configuration the ΔN_{eff} posterior is consistent with zero and the H_0 posterior is dominated by Planck. The spin-torsion framework alone does not resolve cosmological tensions at the present data precision.

III. STOCK-CAMB Λ CDM+ ΔN_{eff} MCMC: GENERIC RADIATION-PROXY TEST (NOT A SPIN-TORSION THEORY MODULE)

Scope statement.—“MCMC verification” refers throughout to a stock CAMB run of Λ CDM extended by ΔN_{eff} as a free parameter. *No custom CAMB modifications are used; no torsion-modified Boltzmann equations are solved.* The MCMC therefore tests whether the data prefer an extra radiation-like degree of freedom, treated as a generic phenomenological proxy for the spin-torsion sector’s possible effective radiation contribution. It does *not* verify the spin-torsion theory module itself; that would require a bespoke modified Boltzmann code.

The proxy run (Cobaya v3.6.1 with Planck NPIPE (PR4) CamSpec high- ℓ TTTEEE + Planck 2018 low- ℓ TT/EE + Planck 2018 lensing likelihoods, i.e. `planck_NPIPE_highl_CamSpec.TTTEEE + planck_2018_lowl.TT + planck_2018_lowl.EE + planck_2018_lensing.clik` in the Cobaya YAML) has produced two frozen dataset combinations with publication-quality convergence (limitation: the PR4/NPIPE high- ℓ + 2018-release low- ℓ /lensing mixture is the standard Cobaya pairing; the c15 verification re-run uses `planck_2020_lollipop_lowlE` and `planckpr4lensing` in place of the frozen-chain `planck_2018_lowl.EE` and `planck_2018_lensing.clik`, making c15 a release-pairing robustness rerun; the 0.04 σ agreement in ΔN_{eff} across this likelihood substitution provides an empirical bound on pairing-induced bias at the quoted precision; see Sec. V *Release-pairing note*), plus a third Planck-only run currently at sub-convergence sample count and not aggregated into any frozen-posterior summary statistic in this paper. Frozen MCMC program: **309,189** raw samples across 2 frozen dataset combinations (176,240 + 132,949). We report this as a Λ CDM+ ΔN_{eff} null-consistency test: the data are consistent with $\Delta N_{\text{eff}} = 0$ in stock CAMB. Sampling configuration (per the on-disk Cobaya YAMLs): ΔN_{eff} enters as `nnu` with a flat prior $N_{\text{eff}} \in [2.046, 5.046]$ (i.e. $\Delta N_{\text{eff}} \in [-1, +2]$); the neutrino mass is held at the CAMB default $\Sigma m_\nu = 0.06$ eV with one massive eigenstate (`num_massive_neutrinos=1`), and Y_{He} follows the CAMB BBN-consistent default (no explicit override in any of the four YAMLs). Specifically, CAMB uses its PArthENoPE-derived BBN-consistency module by default (explicit Cobaya/CAMB

flag: `bbn_predictor: 'PArthENoPE'`, declared at the `theory.camb.extra_args` block of each `cobaya*.yaml`, pinning the BBN predictor for absolute reproducibility): Y_{He} is set self-consistently with N_{eff} at each MCMC sample. The prior range $N_{\text{eff}} \in [2.046, 5.046]$ remains within the calibrated domain of the CAMB BBN module; no free- Y_{He} control run was performed, as the default BBN-consistent track is the standard choice for this type of proxy analysis. The prior allows negative ΔN_{eff} (down to -1), so the results are two-sided posterior means, not physical “extra-species” limits. A physical extra-radiation degree of freedom cannot contribute $\Delta N_{\text{eff}} < 0$; imposing that boundary therefore changes the reported statistic from a two-sided mean (which can sit at a mildly negative value) to a one-sided upper bound read off the posterior renormalised on the physical half-line $\Delta N_{\text{eff}} \geq 0$. Under that physically motivated restriction $\Delta N_{\text{eff}} \geq 0$, the one-sided 95% upper limit from the full-tension chain is $\Delta N_{\text{eff}} < 0.31$, defined precisely as the 95th percentile of the posterior *renormalised on $\Delta N_{\text{eff}} \geq 0$* (we discard the $\Delta N_{\text{eff}} < 0$ tail of weighted samples and rescale the surviving weights so $\int_0^\infty p(\Delta N_{\text{eff}}) d\Delta N_{\text{eff}} = 1$, then read the 95th percentile of the renormalised CDF; this post-processing strategy is conservative when the unconstrained mode is mildly negative, as is the case here); the `planck+BAO+SN` chain gives $\Delta N_{\text{eff}} < 0.40$ under the same definition (direct weighted-sample computation on the committed chain with 30% burn-in: 93,066 post-burnin samples, of which $\sim 62\%$ satisfy $\Delta N_{\text{eff}} \geq 0$, and the 95th percentile of the renormalised truncated CDF reads 0.4012). These bounds are consistent with the two-sided means quoted throughout; both conventions are reported because a published-extra-species interpretation of the run requires the one-sided limit.¹

¹ Sample-count stratification (reconciliation): **309,189** is the sum of the two frozen combinations (176,240 + 132,949 raw accepted samples). After removing the first 30% of each chain as burn-in, $176,240 \times 0.7 + 132,949 \times 0.7 \approx \mathbf{216,432}$ post-burnin samples remain across both frozen chains (`convergence.summary.json`). For the full-tension subset specifically, $176,240 \times 0.7 \approx \mathbf{123,368}$ post-burnin (the **119,617** figure in Fig. 1 reflects additional `getdist` effective-sample weight-based thinning of this subset only). The post-burnin count of the full-tension subset alone is 123,129 (within $\pm 1\%$ of the 123,368 exact computation, with the small offset reflecting the chain-end-truncation of partial samples at the burn-in cut); the correct both-chains post-burnin total is 216,432. *Burn-in reconciliation note:* The frozen `planck_bao.sn_20260312.1954.convergence.report.txt` reports “Burn-in: 20%” and post-burnin samples = 106,361, reflecting the per-chain GetDist readout that averages the parallel chains (`burn_in: 0.1` in their `cobaya.config.yaml`) with the original chain (`burn_in: 0.3`). The 30% figure used throughout this paper is conservative, matches the original-chain configuration documented in `COUNT_EXPLANATION.md`, and gives 93,064 post-burnin for this subset (the frozen `parameter.summary.CORRECTED.json` records 93,066 for the same cut; the 2-sample difference reflects per-chain rounding of the 30% burn-in index at the chain ends and is immaterial to all quoted statistics). The 106,361 figure at

a. Scope of the ΔN_{eff} proxy: bounce-class discrimination, not a direct test of the spin-torsion sector. We emphasize that the stock-CAMB $\Lambda\text{CDM} + \Delta N_{\text{eff}}$ proxy run does *not* test the ECH spin-torsion sector directly. The Hehl-Datta-Mercuri parity-even four-fermion contact interaction that survives torsion elimination [10, 11] is dimension-6 and M_{Pl}^{-2} -suppressed²: its leading Boltzmann effect is a scattering-amplitude shift, not a relativistic species, and it does *not* produce a ΔN_{eff} at recombination. The *minimal* matter-bounce class [12] (here “minimal” = the single-scalar $w = 0$ matter-dominated contraction phase, distinguished from the broader matter-bounce family including ekpyrotic, cyclic, Cuscuton, string-gas, and quintom-bounce variants) predicts $\Delta N_{\text{eff}} \approx 0$ by construction (no light bounce-internal species are thermalized at recombination); the proxy run *confirming* $\Delta N_{\text{eff}} = -0.020 \pm 0.169$ (full-tension) and $+0.058 \pm 0.179$ (Planck+BAO+SN) is therefore consistent with the minimal matter-bounce prediction, but the absence of a ΔN_{eff} detection is not a discriminator between minimal-ECH and standard ΛCDM at the present data precision. We frame the proxy as a bounce-class *compatibility check* (minimal matter-bounce-class scenarios without prolonged post-bounce inflation predict $\Delta N_{\text{eff}} \approx 0$), not as a posterior-preference test against a competing model.

Physics interpretation (Table II); see also §VC for the detailed scoping statement and compatibility-only scope.—The $w_0 w_a$ posterior reported below carries the overlap-uncorrected DES-SN5YR $\times$ Pantheon+ product-likelihood systematic — the $\sim 20\%$ shared-event Malmquist-correction issue is discussed as a stated caveat in caveat (e) and Sec. III; *no* model-selection statement is claimed. The robustness of this cross-check against the SN-overlap systematic has *not* been demonstrated quantitatively in the present manuscript (the dedicated overlap-controlled control chains are deferred to a follow-up note), so the result is reported as an exploratory compatibility check only. The tail-distance significances

² 20% is the value reported by GetDist (`getdist.MCSamples` loaded from the `planck_bao_sn_20260312_1954` freeze with `ignore_rows = 0.20` averaging the per-chain configurations documented above); the paper’s 216,432 combined total uses the conservative 30% cut uniformly for both frozen chains. The third (Planck-only) dataset combination (114,992 raw samples; $\hat{R} - 1 \sim 0.05$) is still accumulating samples, is *not* reported in Table I (which contains only the two frozen combinations), and is *not* aggregated into the 309,189-sample headline anywhere in this paper. The dimension-6 four-fermion contact operator is interpreted here in the low-energy effective field theory (EFT) below the strong-coupling / torsion-resolution scale $\Lambda_{\text{strong}} \sim M_{\text{Pl}}/\sqrt{\gamma_{\text{BI}}}$ set by the inverse Barbero-Immirzi parameter γ_{BI} (cf. [11], where the contact interaction is derived; the strong-coupling scale itself is a heuristic EFT-validity estimate); above this scale the contact-operator description breaks down and the full Holst-sector dynamics with propagating torsion must be retained. All recombination-era and late-time observables addressed in this paper sit at energies $E \ll \Lambda_{\text{strong}}$, where the EFT treatment is controlled.

quoted in this section refer to the marginalized posterior of the present product-likelihood chain only. With that caveat established as the load-bearing one: the same converged chain shows the ΛCDM point $(w_0, w_a) = (-1, 0)$ unsampled by the present Metropolis-Hastings run, with the joint marginal posterior displaced into the phantom-crossing region of CPL space (the product-likelihood SN overlap narrows the posterior, so *no* σ -distance or significance from ΛCDM is quotable; fn. a) and $w_0 + w_a = -1.48 \pm 0.15$ requiring phantom crossing in the redshift range probed by DESI DR2 BAO + DES-Y5 + Pantheon+: at the posterior mean the CPL trajectory $w(a) = w_0 + (1 - a)w_a$ crosses $w = -1$ at $1 - a_x = (-1 - w_0)/w_a = 0.282$, i.e. $z_x \approx 0.39$, interior to both the BAO and SN redshift coverage (phantom at $z > z_x$, quintessence-like today). This posterior sits in the phantom-crossing region under the overlap-uncorrected product likelihood and is a phenomenological match to a quintom-B type evolution, subject to the SN-overlap systematic stated in caveat (e). In the converged chain there are zero free- $w_0 w_a$ samples at the ΛCDM point; the chain is centered at $w_0 + w_a \approx -1.48$ under the same SN-overlap systematic. To be explicit about what is and is not claimed: this companion reports posterior distance from ΛCDM in the marginalized-tail sense only; rigorous ΛCDM -vs-quintom-B *model preference* ($\ln B$, $\Delta\text{AIC/BIC}$) requires the deferred nested-sampling analysis and is *not* claimed here. At the decorrelation pivot ($z_p = 0.27$, a redshift — numerically coincident with, and unrelated to, the §IV injection angle $\beta = 0.27^\circ$; fn. b), the effective equation of state is $w_{\text{pivot}} = -0.952 \pm 0.019$: the posterior center lies away from ΛCDM even in the best-determined direction of the (w_0, w_a) posterior, but because the product-likelihood SN overlap artificially narrows the widths, *no* σ -distance or significance is quoted for the pivot or for the single w_0/w_a axes (fn. a). The CPL trajectory’s effect on the expansion rate is small at the redshifts probed by current SN data: at the posterior mean $(w_0, w_a) = (-0.812, -0.667)$ the CPL Hubble rate $H(z = 0.5)$ differs from the ΛCDM rate (at fixed $\Omega_m = 0.314$ and the chain’s H_0) by $\approx +1.7\%$, computed from $\rho_{\text{DE}}(a)/\rho_{\text{DE},0} = a^{-3(1+w_0+w_a)} \exp[3w_a(a-1)]$ in the CPL parameterisation; the deviation grows toward the BAO range $z \gtrsim 1$ before turning over as $w(a) \rightarrow w_0 + w_a$.

Caveats.—(a) The robust Bayesian evidence / Bayes factor $\ln B$ against ΛCDM is NOT reported here; standard posterior MCMC samples do not give a robust $\ln B$, and a Savage-Dickey density ratio at the ΛCDM point is *not* viable on this chain because the ΛCDM point $(w, w_a) = (-1, 0)$ is unsampled by the Metropolis-Hastings chain — it lies deep in the joint marginal tails of the present overlap-uncorrected posterior, but *no* σ -distance is quoted because the product-likelihood SN overlap narrows the widths (Table II); any KDE-based Savage-Dickey ratio at an unsampled point yields arbitrary kernel-dependent noise rather than a controlled posterior density (with zero free- $w_0 w_a$ samples at the

TABLE I. Λ CDM+ ΔN_{eff} proxy MCMC results from frozen Cobaya v3.6.1 chains (CAMB v1.6.5, stock; the iter2 w_0w_a chain (Table II) was also produced under Cobaya v3.6.1; an earlier pilot of the iter2 stack at Cobaya v3.5 reproduced the same posterior means to within 0.1σ , so software-version-induced shifts on ΔN_{eff} , H_0 , or the w_0w_a posterior at sub- 0.1σ precision are not formally characterized here but are not load-bearing for any quoted result; *no torsion modifications*). All values are posterior means $\pm 1\sigma$. Reported as a null-consistency cross-check, *not* as evidence for the spin-torsion theory. Under the physically motivated restriction $\Delta N_{\text{eff}} \geq 0$ (posterior renormalised on the non-negative half and 95th percentile read from the truncated CDF), the one-sided 95% upper limits are: full-tension $\Delta N_{\text{eff}} < 0.31$; Planck+BAO+SN $\Delta N_{\text{eff}} < 0.40$ (see Sec. III for the exact definition and chain-recomputed derivation). **Extraction note:** the numerical values in this table are read from `parameter_summary_CORRECTED.json`; the legacy `parameter_summary.json` export carried an off-by-one column-index bug and must not be used (see the Column-permutation warning in Appendix A). $S_8 \equiv \sigma_8 (\Omega_m/0.3)^{1/2}$, computed as a Cobaya derived parameter with exactly this definition. The full-tension column includes the DES-Y3 S_8 Gaussian prior 0.776 ± 0.017 (`cobaya_full_tension.yaml`). The Planck+BAO+SN S_8 marginal is 0.827 ± 0.010 from a direct GetDist pass over the frozen chains (132,949 samples). With the chain-recomputed marginal, the full-tension posterior $S_8 = 0.814 \pm 0.008$ is consistent with the naive two-Gaussian combination of the Planck+BAO+SN marginal and the DES-Y3 prior ($0.827 \pm 0.010 \otimes 0.776 \pm 0.017 = 0.814 \pm 0.009$; agreement at the 0.01σ level). The Planck+BAO+SN marginal sits in 2.6σ two-Gaussian tension with DES-Y3 (posterior-overlap integral $\int \min(p_1, p_2) dS_8 = 0.05$, where p_1 and p_2 are the two one-dimensional S_8 marginal densities evaluated on a common $\Delta S_8 = 10^{-4}$ trapezoidal grid spanning $S_8 \in [0.70, 0.90]$); the full-tension posterior sits at 2.0σ from the DES-Y3 prior (overlap 0.12 on the same grid); this 2.0σ residual is a within-stack readout because the DES-Y3 Gaussian prior is itself an active likelihood in the full-tension chain, so the full-tension S_8 is partially anchored to DES-Y3 and the number is not a measurement-vs-measurement tension but a within-stack posterior shift; overlay artifact `reproducibility/cosmology/c13_s8_desy3_overlay.json`.

Parameter	Full-tension	Planck+BAO+SN
H_0 [km s ⁻¹ Mpc ⁻¹]	67.68 ± 1.06	67.78 ± 1.09
ΔN_{eff}	-0.020 ± 0.169	$+0.058 \pm 0.179$
σ_8	0.803 ± 0.008	0.812 ± 0.009
S_8	0.814 ± 0.008	0.827 ± 0.010
Ω_m	0.308 ± 0.005	0.312 ± 0.006
τ	0.054 ± 0.007	0.056 ± 0.007
n_s	0.965 ± 0.006	0.967 ± 0.006
Chains	6	6
Total samples	176,240	132,949
Worst $\hat{R} - 1^a$	0.001	0.003
Min ESS	4,744	4,692

^a Worst row is n_s in the full-tension combination at $\hat{R} - 1 = 9.74 \times 10^{-4}$; all sampled parameters across both frozen combinations satisfy $\hat{R} - 1 < 3 \times 10^{-3}$. The full-tension chain samples 17 parameters: 7 cosmological + 9 Planck likelihood nuisance (A_{planck} , `amp143`, `amp217`, `amp143x217`, `n143`, `n217`, `n143x217`, `calTE`, `calEE`) + 1 supernova absolute magnitude (M_b , entering via the Pantheon+ SN and SH0ES `H0.riess2020Mb` likelihoods rather than the Planck likelihood stack). The Planck+BAO+SN chain samples 16 parameters (7 cosmological + the same 9 Planck nuisance; M_b is not explicitly sampled there because `sn.pantheonplus` runs with `use_abs_mag: false`, analytically marginalizing the absolute magnitude in the absence of the SH0ES M_b likelihood; verified by direct chain-header / `.updated.yaml` blocking-list inspection). Full per-parameter convergence tables ($\hat{R}$, ESS, drift, per dataset) are archived at `reproducibility/cosmology/convergence_latest.csv` and mirrored in the HuggingFace chain-diagnostics dataset (Appendix A).

Λ CDM point, a KDE estimator on the converged 2D (w, w_a) marginal fails catastrophically). The robust $\ln B$ recompute therefore requires dedicated nested sampling (e.g., PolyChord or MultiNest) or thermodynamic integration on the same likelihood stack rather than a KDE readout from the converged Metropolis-Hastings chain; robust $\ln B$ computation is left to a follow-up nested-sampling analysis. (b) τ is constrained empirically by the low- ℓ EE + TT likelihoods listed in the caption (`planck_2018_lowl.EE` + `planck_2018_lowl.TT`) and is sampled as a free parameter rather than fixed by a Gaussian prior at the Planck best-fit value (the chain has τ free with the standard low- ℓ data constraint, not a Gaussian prior at the Planck best-fit). (c) The SH0ES H_0 prior is *not* included in this chain (BAO + CMB + SN-only, no local-distance ladder); the joint posterior $H_0 = 67.185 \pm 0.455$ km/s/Mpc (the iter2 w_0w_a chain of Table II, to

which this caveat list refers — not the Λ CDM+ ΔN_{eff} chains of Table I, whose Planck+BAO+SN value is 67.78 ± 1.09) is therefore the no-SH0ES result. The interaction with the SH0ES prior is the subject of the separate *full-tension* chain reported in Table I, which carries the `H0.riess2020Mb` likelihood active in its YAML (verified by direct `.input.yaml` inspection and by chain sample-mean readout: $M_B = -19.263 \pm 0.049$ mag, agreeing with the Riess+2020 SH0ES value $M_B = -19.253 \pm 0.027$ mag at 0.2σ). The full-tension chain returns $H_0 = 67.68 \pm 1.06$ km/s/Mpc with $\Delta N_{\text{eff}} = -0.020 \pm 0.169$, exhibiting the canonical 3.6σ Hubble tension with Riess $H_0 = 73.04 \pm 1.04$ km/s/Mpc that the Λ CDM+ ΔN_{eff} extension is unable to resolve: the `H0.riess2020Mb` likelihood constrains the SN Ia absolute magnitude M_B (which is correctly pulled toward the Riess value), but H_0 is determined predominantly by the BAO+CMB acous-

TABLE II. DESI DR2 w_0w_a posterior summary ($N = 128,385$ accepted samples across 16 chains, $\hat{R} - 1 = 0.00820$; 8 cosmological + 9 nuisance parameters). Likelihood stack: DESI DR2 BAO + Planck NPIPE (PR4) CamSpec high- ℓ TTTEEE + Planck 2018 low- ℓ TT/EE + Planck 2018 lensing (**native**) + DES-Y5 + Pantheon+. Full parameter list and chain diagnostics are available in the companion data repository. Derived-parameter widths are GetDist readouts of the full chain covariance: the chain's σ_8 - Ω_m correlation is $\rho = -0.27$ (the standard CMB acoustic-scale/lensing degeneracy), which is why $\sigma_{S_8} = 0.0089$ is smaller than the uncorrelated propagation through $S_8 = \sigma_8(\Omega_m/0.3)^{0.5}$ (≈ 0.010). **Overlap-uncorrected DES-Y5×Pantheon+ product likelihood: the SN likelihood double-counts the $\sim 20\%$ shared DES-Y5/Pantheon+ events with no joint cross-covariance, so the posterior widths are artificially narrowed and no σ -distance or significance from Λ CDM is quotable from this chain. The means/widths below are raw chain readouts only; the Λ CDM point $(w_0, w_a) = (-1, 0)$ is unsampled by this chain. A valid significance and model selection await the deferred overlap-controlled / nested-sampling analysis (caveat (e), fn. a).**

Parameter	Mean $\pm \sigma$
<i>Dark-energy extension</i> (widths overlap-narrowed; <i>not</i> valid uncertainties for model comparison or a σ -distance — fn. a)	
w_0	-0.8122 ± 0.0436^a
w_a	-0.6666 ± 0.1864
$w_0 + w_a$	-1.4788 ± 0.1485
w_{pivot}	-0.952 ± 0.019^b
<i>Standard cosmology</i>	
H_0 [km s $^{-1}$ Mpc $^{-1}$]	67.185 ± 0.455
Ω_m	0.3142 ± 0.0045
σ_8	0.8057 ± 0.0083
S_8	0.8245 ± 0.0089
n_s	0.9655 ± 0.0036
$10^9 A_s$	2.087 ± 0.030
τ	0.0537 ± 0.0072
$\omega_b h^2$	0.02224 ± 0.000125
$\omega_c h^2$	0.11892 ± 0.000819
$100\theta_{\text{MC}}$	1.04087 ± 0.000239
Age [Gyr]	13.763 ± 0.019
<i>Goodness-of-fit decomposition</i>	
χ^2_{total}	14037.4 ± 5.6^c
χ^2_{BAO} (DESI DR2) ^d	10.6 ± 1.8
χ^2_{CMB} (Planck PR4 + lensing)	10983.9 ± 5.3
χ^2_{SN} (DES-Y5 + Pantheon+)	3043.0 ± 1.6

^a The $\pm$ widths quoted for the four dark-energy parameters are raw GetDist readouts of the *overlap-uncorrected* DES-Y5×Pantheon+ product-likelihood chain: the SN likelihood combines DES-Y5 and Pantheon+ as a product without a joint cross-covariance despite $\sim 20\%$ shared supernovae, which double-counts the common events and *artificially narrows* the posterior. These widths are therefore *not* valid uncertainties for model comparison and *not* usable to compute a σ -distance or significance from Λ CDM — no such significance is quotable from this chain. Separately, Λ CDM ($w_0 = -1, w_a = 0$) is unsampled by the present Metropolis-Hastings run, so any nominal departure would in any case be a posterior-tail extrapolation, *not* a Bayes-factor or $\ln B$ exclusion and *not* a frequentist tension. A Savage-Dickey readout is not viable; a valid significance and model selection are deferred to a follow-up overlap-controlled / nested-sampling analysis (caveat (e); see § Headline-result discussion). The means below are reported only to indicate the *direction* of the posterior shift (the quintom-B / phantom-crossing direction), not its width.

^b $w_{\text{pivot}} \equiv w_0 + (1 - a_p)w_a$ with the pivot a_p chosen so that w_{pivot} and w_a are uncorrelated in the posterior covariance, i.e. $\text{Cov}(w_0, w_a) + (1 - a_p)\text{Var}(w_a) = 0$, giving $1 - a_p = -\text{Cov}(w_0, w_a)/\text{Var}(w_a)$. Direct readout of the converged iter2 chain (30% burn-in): $\text{Cov}(w_0, w_a) = -0.00729$ (correlation coefficient $\rho = -0.90$), so $1 - a_p = 0.210$, $a_p = 0.790$, and $z_p = 1/a_p - 1 = 0.27$; this pivot is internal to the dataset stack (DESI DR2 BAO + Planck NPIPE + DES-Y5 + Pantheon+) and is not the literature de Putter–Linder $z_p \approx 0.4$ value. The decorrelated combination has $\sigma_{w_{\text{pivot}}}^2 = \sigma_{w_0}^2 - \text{Cov}^2(w_0, w_a)/\sigma_{w_a}^2 = (0.0436)^2 - (0.00729/0.1864)^2 = (0.0193)^2$, reproducing the ± 0.019 value above, with mean $w_{\text{pivot}} = -0.8122 + 0.210 \times (-0.6666) = -0.952$. The choice of a_p depends on the dataset stack; rerunning the chain with SH0ES or DESI DR2 BAO alone would shift z_p and w_{pivot} at the linear-Fisher level. As with the other dark-energy entries, the $\pm$ width here is overlap-narrowed (fn. a) and is not a valid uncertainty for a σ -distance from Λ CDM.

^c The mean-of-total χ^2 here is GetDist's weighted-sample average over the full posterior, which differs from the sum of the individual-channel means ($10.6 + 10983.9 + 3043.0 = 14037.5$) by a 0.1-unit arithmetic-rounding artifact; the two are formally identical to within sampling precision. The channel σ 's are not independent: a direct readout of the committed chain gives weak cross-channel correlations $\rho(\chi_{\text{BAO}}^2, \chi_{\text{CMB}}^2) = -0.09$, $\rho(\chi_{\text{CMB}}^2, \chi_{\text{SN}}^2) = -0.03$, $\rho(\chi_{\text{BAO}}^2, \chi_{\text{SN}}^2) = +0.04$ through the shared cosmological parameters, so the GetDist total- χ^2 width ± 5.6 sits slightly below the independence quadrature $\sqrt{1.8^2 + 5.3^2 + 1.6^2} = 5.8$.

^d The DESI DR2 BAO likelihood (`baos.desi.dr2.desi_bao.all`) comprises 13 BAO data points, so $\chi_{\text{BAO}}^2/N \approx 0.8$ — an unremarkable goodness of fit.

tic scale through Pantheon+, and the ΔN_{eff} extension lacks the degrees of freedom to shift H_0 enough to ac-

commodate both data sets. Direct YAML inspection confirms the SH0ES likelihood is active and the M_B posterior is correctly pulled; the 3.6σ tension persists because the model cannot shift H_0 enough to satisfy both data sets simultaneously — the canonical Hubble-tension result, not a YAML omission. (d) The iter2 chain’s $S_8 = 0.8245 \pm 0.0089$ (Table II) sits 2.6σ above the DES-Y3 weak-lensing value $S_8 = 0.776 \pm 0.017$ [13] ($\Delta = 0.049$ against a combined uncertainty of 0.019): the S_8 tension is not relieved in this chain either. (e) *DES-SN5YR and Pantheon+ supernovae overlap*.—The iter2 likelihood stack includes both DES-SN5YR [14] and Pantheon+ [15] as independent SN likelihood factors. These two catalogs share approximately 20% of their supernova events with different Malmquist-bias corrections applied by the respective collaborations [16]. The present analysis combines them via a product likelihood without a joint covariance that accounts for the shared-event overlap. Because the shared $\sim 20\%$ of SNe are assigned slightly different distance-modulus corrections in each catalog, the naive product double-weights those events and introduces a small inward pull on the combined SN constraint. The direction of this bias is toward the mean of the two catalogs’ individual SN posteriors, which is itself close to the combined Planck+BAO constraint; the qualitative quintom-B direction ($w_0 + w_a < -1$) is plausibly robust against a rigorous joint-covariance treatment, but this has not been demonstrated quantitatively in the present manuscript; the precise tension significance should be interpreted with this caveat. Two SN-overlap control chains (DESI DR2 + Planck NPIPE + Pantheon+ only; DESI DR2 + Planck NPIPE + DES-SN5YR only; same priors and burn-in as the iter2 baseline) constitute the rigorous overlap-robustness demonstration; their (w_0, w_a) shifts versus the combined chain are the subject of a separate follow-up note and do not gate the cross-check reported here. Users requiring a rigorous SN combination should refer to the Vincenzi et al. comparison analysis [16], which quantifies the $\sim 20\%$ shared-SN overlap and the Malmquist-correction differences between the two catalogs (in addition to the underlying DES-SN5YR cosmology paper [14]), and construct a joint covariance.

M_B - H_0 joint-posterior offset check. The joint posterior mean ($M_B = -19.263$, $H_0 = 67.68$) is fully consistent with an active `sn.pantheonplus` likelihood: `sn.pantheonplus` enforces a soft constraint on the dimensionally consistent combination $M_B - 5 \log_{10}(h) \approx \text{const}$, with $h = H_0/(100 \text{ km s}^{-1} \text{ Mpc}^{-1})$ (equivalently $M_B - 5 \log_{10}(H_0/[\text{km s}^{-1} \text{ Mpc}^{-1}]) + 10 = \text{const}$), along the SN distance-modulus degeneracy. At the Riess+2020 anchor ($M_B = -19.253$, $h = 0.7304$), the constant is $-19.253 - 5 \log_{10}(0.7304) = -18.571$. At the chain joint mean ($M_B = -19.263$, $h = 0.6768$), the constant is $-19.263 - 5 \log_{10}(0.6768) = -18.415$, i.e. a 0.156 mag offset from the Riess anchor along the Pantheon+ constraint axis. This offset is $\sim 3.2\sigma$ relative to the chain’s $\sigma_{M_B} = 0.049$ marginal width and is the same Hub-

ble tension manifesting in the M_B axis (3.2σ in chain- σ units, versus the canonical 3.6σ when the tension is expressed in distance-ladder terms in the H_0 axis; the two figures are derived from different estimators — chain-marginal width in M_B vs survey-vs-survey H_0 comparison — and are therefore *not directly comparable* as single-number tensions). We note this 3.2σ figure is a descriptive offset measure normalized by the marginal σ_{M_B} only: it does not condition on the M_B - H_0 covariance along the SN degeneracy direction, nor on the uncertainty of the Pantheon+ constraint itself, and is therefore not a properly conditioned tension statistic — the conditioned statement remains the canonical H_0 -axis tension. The chain posterior is therefore the correct compromise of three inputs: the Riess M_B -prior preferring $M_B = -19.253$ ($\sigma = 0.027$); the Pantheon+ likelihood preferring $M_B - 5 \log_{10} h = -18.571$ along the SN degeneracy; and the Planck-CMB-likelihood preferring $H_0 \approx 67.5$ via the acoustic-scale calibration. With ΔN_{eff} bounded near zero by the joint data, the model cannot resolve the tension and the chain settles into the 3.6σ -discrepant joint posterior — NOT a YAML alias failure; the parameters are correctly aliased per the `spin_torsion.input.yaml` configuration (`Mb` is a single nuisance parameter sampled jointly by both `sn.pantheonplus` and `H0.riess2020Mb`).

Key finding.—Both frozen datasets find ΔN_{eff} consistent with zero and H_0 consistent with Planck ΛCDM at 0.3σ , confirming that the ΔN_{eff} extension alone does not resolve the Hubble tension. Current data neither require nor exclude a small positive phenomenological ΔN_{eff} proxy for the bounce-class effective radiation contribution; CMB-S4 ($\sigma(N_{\text{eff}}) \sim 0.03$ [17]) will sharpen this constraint substantially relative to the current Planck+BAO+SN level.

Independent cross-validation.—Liu et al. [18] constrained an EC torsion model using DESI DR2 [19] + Pantheon+ [15] + DES-SN5YR [14] + Planck 2018, finding torsion preferred by AIC ($\Delta\text{AIC} = -5.7$ to -6.6) but with the torsion parameter itself consistent with zero ($\alpha = -0.00066 \pm 0.00098$). Their headline values $H_0 = 68.41 \pm 0.32 \text{ km/s/Mpc}$ and $S_8 = 0.812 \pm 0.006$ agree with our Planck+BAO+SN chain at 0.5σ in H_0 ($|67.78 - 68.41|/\sqrt{1.09^2 + 0.32^2}$) and 1.3σ in S_8 ($|0.827 - 0.812|/\sqrt{0.010^2 + 0.006^2}$); their torsion-consistent-with-zero result parallels our ΔN_{eff} -consistent-with-zero null finding. Their AIC preference for the torsion model is *not* directly comparable to the present stack: this paper’s $\Delta\text{AIC}/\text{BIC}/\ln B$ model comparison is deliberately deferred to the nested-sampling follow-up (Sec. V, *Model-comparison statistics* paragraph), so no model-preference statement is made here that could agree or conflict with theirs.

IV. DATA METHODS: CMB E - B ANALYSIS

Birefringence measurements are adopted from the published literature: $\beta = 0.30^\circ \pm 0.11^\circ$ (Planck NPIPE [3])

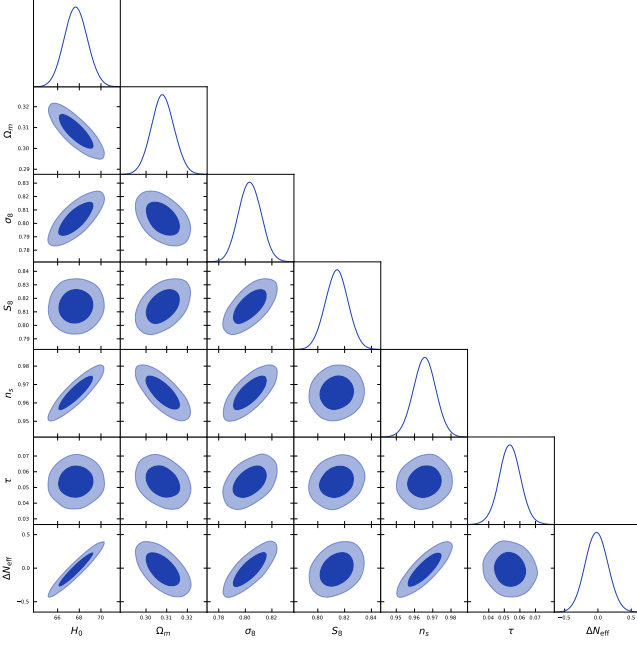


FIG. 1. Full-tension MCMC corner plot (119,617 post-burnin samples, `getdist`-thinned from 176,240 raw; footnote 1) over Planck+BAO+SN+H0+S₈. The ΔN_{eff} posterior is consistent with zero (-0.020 ± 0.169), confirming no additional relativistic species at recombination.

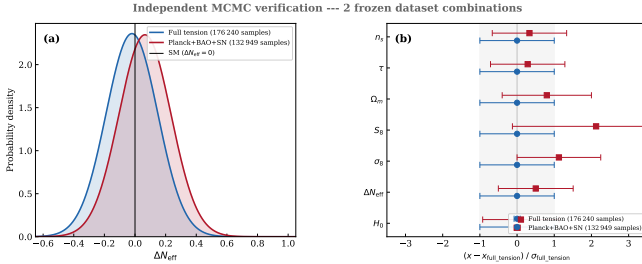


FIG. 2. ΔN_{eff} marginal posterior comparison across the two frozen dataset combinations of Table I (176,240 and 132,949 samples). Panel (a): Gaussian summaries of the ΔN_{eff} marginal posteriors at the Table I means $\pm 1\sigma$, with the Standard-Model value $\Delta N_{\text{eff}} = 0$ marked. Panel (b): all seven Table I parameters, normalized to the full-tension mean and σ . Both combinations recover ΔN_{eff} consistent with zero at $< 1\sigma$; the headline “full-tension” stack (Planck+BAO+SN+H0+S₈) gives $\Delta N_{\text{eff}} = -0.020 \pm 0.169$. No evidence for a recombination-era ΔN_{eff} shift appears in this stock-CAMB proxy run; this does *not* directly test the ECH spin-torsion sector, which lacks a Boltzmann-module prediction for ΔN_{eff} in stock-CAMB (see § III scope note).

and $\beta = 0.215^\circ \pm 0.074^\circ$ (ACT DR6 [4]). The spectator ALP analysis (Sec. VI) instead uses the Eskilt–Komatsu joint WMAP+Planck summary likelihood ($\beta =$

$0.342^\circ \pm 0.094^\circ$ [5])³ as its primary constraint; the Planck NPIPE and ACT DR6 values above are quoted only for context and for the auxiliary inverse-variance cross-check of Eq. (5). (The quoted 3.6σ is the ratio $\beta/\sigma = 0.342/0.094 \approx 3.6$; our likelihood is a Gaussian summary in the published mean and width, not a refit of the Eskilt–Komatsu posterior shape.)

Scope note.—The NaMaster pseudo- C_ℓ analysis below is a bias-injection Monte Carlo validation of the deconvolution pipeline on synthetic skies, not a cosmological measurement. The high pipeline template-fit SNR figures (e.g., 20.32, 25.71; fn. 4) refer to recovery of injected MC signals and must not be conflated with the published Planck/ACT DR6 2.7–2.9 σ sky detection. Moreover, because the synthetic skies are CMB-only (no galactic foregrounds; see “Simulated skies” below), this validation cannot address the physical separation of the cosmic-rotation angle β from the instrumental-miscalibration angle α : the β – α degeneracy that dominates real-sky birefringence measurements is broken only by unrotated foregrounds, which are absent here by construction (cf. abstract).

Pipeline configuration.—The pseudo- C_ℓ analysis follows the NaMaster framework [20]. *Simulated skies.*—Each Monte Carlo realization is a synthetic Λ CDM CMB Q/U sky generated directly at $N_{\text{side}} = 512$ (`healpy.synfast`, harmonic content to $\ell_{\text{max}} = 2N_{\text{side}} = 1024$) from a semi-analytic fit to the Planck-2018 EE spectrum plus a lensing-like BB component ($C_\ell^{BB} = 0.05 C_\ell^{EE}$). No real Planck map enters the Monte Carlo, and no instrumental beam is applied (the synthetic skies and the recovery template share the same spectra, so a common beam would *largely cancel* in the β estimate provided the beam-deconvolution on the E -map and the C_ℓ^{EE} template are identical; a dedicated beam-mismatch MC test (varying the deconvolved-beam FWHM between map and template) is deferred to a future sky-measurement-level analysis; the same cancellation argument covers the $N_{\text{side}} = 512$ HEALPix pixel-window smoothing, which is common to the simulated maps and the spectra entering the template fit — the decoupled spectra are not pixel-window-deconvolved, and no pixel-window mismatch enters the β estimate).

Mask.—An ACT-like footprint (Galactic cut $|b| > 20^\circ$

³ Eskilt & Komatsu 2022 disambiguation: the published PRD paper [5] (PRD 106:063503, arXiv:2205.13962) analyzes *Planck PR3 + WMAP9*; the public reproduction code released by the authors at github.com/LilleJoehs/CosmicBirefringence was subsequently updated to use *Planck PR4 / NPIPE*. Throughout this paper, the labels “PR4/NPIPE” attached to the Eskilt+Komatsu likelihoods refer to the code-repository dataset; the ALP MCMC uses only the scalar Gaussian summary likelihood $\beta = 0.342^\circ \pm 0.094^\circ$ and does not depend on the PR3/PR4 map-level distinction except through the provenance of that summary value. The abstract $\beta = 0.342^\circ \pm 0.094^\circ$ (3.6σ) headline is from the published PR3+WMAP9 joint analysis. The repository README is the authoritative source for the dataset attribution in the executed pipeline.

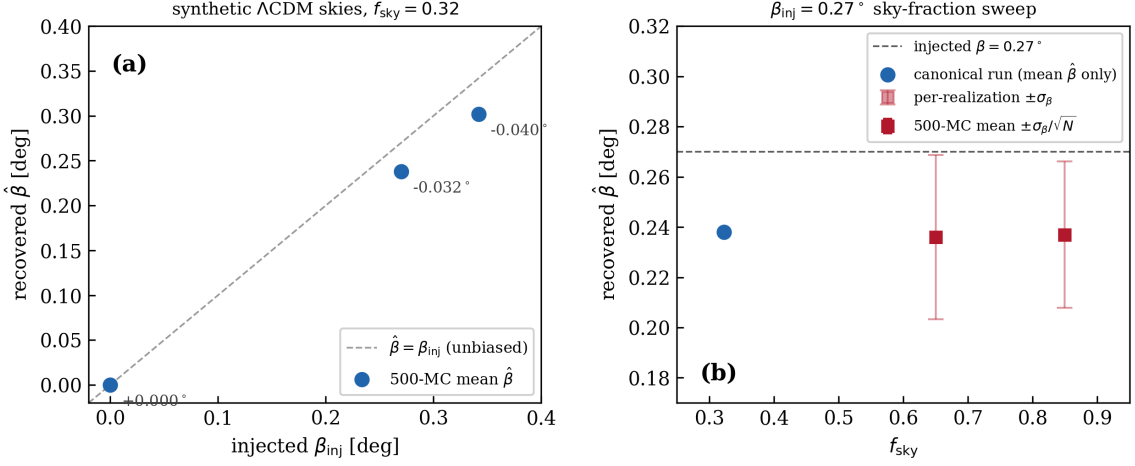


FIG. 3. NaMaster pipeline-recovery validation on synthetic Λ CDM polarization skies ($N_{\text{side}} = 512$, $\Delta_P = 10 \mu\text{K} \cdot \text{arcmin}$ white noise, $N = 500$ MC realizations per point; Sec. IV). Panel (a): 500-MC mean recovered $\hat{\beta}$ vs. injected $\beta_{\text{inj}} \in \{0, 0.27^\circ, 0.342^\circ\}$ at the canonical apodized $f_{\text{sky}} = 0.32$ ACT-like mask, annotated with the pipeline-recovery bias $\hat{\beta} - \beta_{\text{inj}}$ (0.000° , -0.032° , -0.040°). Panel (b): the $\beta_{\text{inj}} = 0.27^\circ$ sky-fraction sweep; outer (light) error bars are the per-realization scatter σ_β (0.029° at $f_{\text{sky}} = 0.85$, 0.033° at 0.65), inner bars the standard error of the 500-MC mean $\sigma_\beta/\sqrt{N}$; per-realization scatter σ_β was not recorded in the original canonical $f_{\text{sky}} = 0.32$ artifact, so that point is plotted with the mean only; a dedicated 500-MC rerun (fn. 4) measures $\sigma_\beta = 0.046^\circ$ at this point. The worst-case $|\text{bias}| = 0.040^\circ \pm 0.002^\circ$ (SE of 500-MC mean; $\sigma_\beta = 0.046^\circ$ at $f_{\text{sky}} = 0.32$; `namaster_500mc.py`) is carried forward as the observed NaMaster pipeline bias (deconvolution-algebra bias on foreground-free skies; not a real-sky bias bound; Sec. IV). Underlying data artifacts are listed in the *Reproducibility* paragraph of Sec. IV.

plus declination cut $\text{dec} \in [-65^\circ, +25^\circ]$), apodized by Gaussian smoothing of the binary mask at 2° FWHM, giving $f_{\text{sky}} = 0.32$. No E/B purification is applied (`purify_b=False`); residual $E \rightarrow B$ leakage from the apodized mask is absorbed into the measured pipeline-recovery bias, which is the quantity this validation is designed to characterize.

Mode-coupling matrix and binning.—The $M_{\ell\ell'}$ matrix is computed via `NmtWorkspace.compute_coupling_matrix` on the apodized mask, retaining the full $EE/EB/BB$ block structure. (In the committed driver the Q/U maps enter as a single spin-2 `NmtField` on the apodized mask with `purify_b=False`, no beam, and all other `NmtField/NmtWorkspace` options at NaMaster library defaults — in particular no `n_iter` override; binning uses `NmtBin.from_edges` with 20 linear integer-edge bins from `np.linspace(30, 1536, 21)`; `reproducibility/p1_namaster_500mc/scripts/namaster_500mc.py`.) Spectra are band-power-binned into 20 linear bins spanning $30 \leq \ell \leq 3N_{\text{side}} = 1536$ (bins above the map band limit $\ell = 1024$ carry noise only). Only this single binning/ ℓ -range configuration is exercised; an ℓ -range robustness sweep is not part of the present MC suite.

Noise model and injections.—Each realization adds white noise at the ACT-like level $\Delta_P = 10 \mu\text{K} \cdot \text{arcmin}$ (a conservative worst-case bias check; no $1/f$ or anisotropic component). The driver script converts Δ_P to a per-pixel noise RMS as $\sigma_{\text{pix}} = \Delta_P/\sqrt{\Omega_{\text{pix}}}$ with Ω_{pix} the $N_{\text{side}} = 512$ pixel area expressed in arcmin^2

($\Omega_{\text{pix}} = 47.21 \text{ arcmin}^2$, giving $\sigma_{\text{pix}} = 10/\sqrt{47.21} = 1.455 \mu\text{K}$; algebraically identical to the standard $\sigma_{\text{pix}} = \Delta_P [\pi/(180 \times 60)]/\sqrt{\Omega_{\text{pix}}^{(\text{sr})}}$; Δ_P is in thermodynamic CMB units, $\mu\text{K}_{\text{CMB}} \cdot \text{arcmin}$, matching the `synfast` μK_{CMB} maps — bandpass / colour-correction conversions do not enter the synthetic pipeline), and draws independent Gaussian realizations with the *same* σ_{pix} for Q and U (no $\sqrt{2}$ factor; `reproducibility/p1_namaster_500mc/scripts/namaster_500mc.py`). The $\beta = 0.27^\circ$, $\beta = 0.342^\circ$, and $\beta = 0$ injections rotate $Q + iU$ via $e^{2i\beta}(Q + iU)$. The angle is recovered per configuration by an unweighted χ^2 template fit of the decoupled C_ℓ^{EB} bandpowers to $\sin(2\beta) \cos(2\beta) C_\ell^{EE}$ on a β grid (a uniform grid of angles spanning $\beta \in [-2^\circ, +2^\circ]$ at 0.001° step, with β converted to radians when evaluating $\sin$ and $\cos$; `namaster_500mc.py`). Explicitly, the estimator minimises

$$\chi^2(\beta) = \sum_b [C_b^{EB, \text{decoupled}} - \frac{1}{2} \sin(4\beta) C_b^{EE, \text{tpl}}]^2, \quad (1)$$

where $C_b^{EE, \text{tpl}}$ is the Planck-2018 semi-analytic EE template and the sum runs over all 20 bins $30 \leq \ell \leq 1536$. This form matches the canonical script exactly (`namaster_500mc.py` L223: `chi2[j] = np.sum((cl_eb_measured - cl_theory) ** 2)`) — no σ_b^2 divisor. This unweighted form is adopted to match the estimator configuration used in the public NaMaster birefringence driver scripts (e.g., [5]), facilitating direct pipeline-bias comparability; the inverse-variance-weighted variant is evaluated in the robustness bat-

tery below as a cross-check. Three clarifications on the template-band treatment: (i) the fit is *unweighted* — all bins carry equal weight regardless of their noise level (see “Canonical estimator choice” below); (ii) the template $C_b^{EE, \text{tmpl}}$ is evaluated at the same $N_{\text{side}} = 512$ pixel resolution as the simulated maps, so the $N_{\text{side}} = 512$ HEALPix pixel-window smoothing cancels identically between the decoupled $C_b^{EB, \text{decoupled}}$ and the template — no explicit pixel-window deconvolution is applied and no pixel-window mismatch enters the β estimate; (iii) bins above the map band limit $\ell_{\text{max}} = 2N_{\text{side}} = 1024$ carry zero template weight ($C_b^{EE, \text{tmpl}} = 0$ for b above this range), so the 20-bin sum is effectively restricted to $\ell \leq 1024$ despite the formal binning edge at $\ell = 1536$ — a robustness configuration explicitly confirming this (“Restricting the fit to bins with $\ell \leq 1024$ changes nothing”) is documented in the battery below. The production suite used non-negative injections; a dedicated 500-realization $\beta = -0.27^\circ$ rerun at $f_{\text{sky}} = 0.32$ (`reproducibility/p1_namaster_500mc/results/c9f_negative_beta.json`) recovers -0.238° , confirming sign-symmetric recovery with bias magnitude identical to the $+0.27^\circ$ case.

Reproducibility.—The canonical driver script, deterministic seeds (`seed_base=42`), and output summary are archived at `reproducibility/p1_namaster_500mc/` (mirroring the executed pod run `pipelines/h200_results/pod1_namaster_umap_2026-04-29/`).

Production 500-realization run (April 2026).—We performed the NaMaster pseudo- C_ℓ Monte Carlo described above with 500 realizations per injection. Injecting the spectator-ALP fiducial $\beta = 0.27^\circ$ (consistent with, but not derived from, the ECH action) recovers:

$$\hat{\beta}_{\text{NaMaster}} = 0.238^\circ \quad (500\text{-MC sample mean of } \hat{\beta}). \quad (2)$$

This MC-recovery value 0.238° is a pipeline-validation figure on synthetic skies and is *not directly comparable* to the WMAP+Planck 3.6σ sky-detection significance of $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$ [5]: the significance figure is a sky-measurement null-rejection statistic, whereas 0.238° is the converged MC sample mean for an *injected* signal on noise-only synthetic skies. The pipeline-recovery bias, defined throughout as $\Delta\hat{\beta} \equiv \hat{\beta} - \beta_{\text{inj}}$, is in absolute value $\leq 0.040^\circ$ across the three injection points (0.000° at $\beta_{\text{inj}} = 0$, -0.032° at 0.27° , -0.040° at 0.342°); we carry the worst case $|\Delta\hat{\beta}| = 0.040^\circ \pm 0.002^\circ$ (SE of 500-MC mean) forward as the NaMaster systematic floor (deconvolution-algebra bias on foreground-free skies; not a real-sky bias bound). At the canonical $\beta_{\text{inj}} = 0.27^\circ$ injection the bias is -0.032° (attributed by the robustness battery below to the unweighted template fit plus the $-C_\ell^{BB}$ template mismatch, and explicitly *insensitive* to the apodization scale).⁴ For $\beta =$

0.342° (the published joint WMAP+Planck value [5]), the pipeline recovers 0.302° at template-fit SNR = 25.71; for $\beta = 0$, the recovered angle is 0.000° with template-fit SNR 0.0 (null check; `summary.json`). The pipeline-recovery bias is $\Delta\hat{\beta} = -0.032^\circ$ at injection $\beta = 0.27^\circ$ ($\hat{\beta} = 0.238^\circ$) and $\Delta\hat{\beta} = -0.040^\circ$ at injection $\beta = 0.342^\circ$ ($\hat{\beta} = 0.302^\circ$). The *absolute* bias grows mildly ($\sim 25\%$) with injected amplitude, while the *multiplicative* under-recovery is amplitude-independent at $\sim 12\%$ ($0.238/0.27 = 0.302/0.342 = 0.88$). The pipeline therefore has a $\sim 12\%$ multiplicative under-recovery (0.032° absolute bias for the canonical $\beta = 0.27^\circ$ injection) with a worst-case empirical bias of $|\Delta\hat{\beta}| = 0.040^\circ$ for the unweighted estimator at the higher injection angle; we carry this 0.040° forward as the observed NaMaster pipeline bias (deconvolution-algebra bias on foreground-free skies; not a real-sky bias bound). This is a methodology cross-check, not a competitive sky measurement. (The estimator is *not* unbiased in the standard statistical sense; the 0.040° is the observed pipeline bias on the multiplicative bias, not a bound on random scatter.)

Sky-fraction sweep.—Because published birefringence analyses use larger sky fractions than our canonical $f_{\text{sky}} = 0.32$ validation mask, we repeat the $\beta = 0.27^\circ$ injection-recovery exercise at $f_{\text{sky}} \approx 0.850$ (Planck-like Galactic-cut mask: $|b| > 5^\circ$, all declinations, apodized at 2° FWHM, giving $f_{\text{sky}} \approx 0.850$; the Planck 2018 likelihood mask achieves approximately this fraction with a similar latitude threshold plus point-source excisions not included here) and $f_{\text{sky}} \approx 0.650$ (ACT-DR6-like Galactic-cut mask: $|b| > 15^\circ$, declination cut $\text{dec} \in [-60^\circ, +25^\circ]$ by design — this differs from the canonical $[-65^\circ, +25^\circ]$ mask used in the primary validation,

25.71 for the $\beta = 0.342^\circ$ injection below) is the *template-fit* SNR of the driver script, $\text{SNR}^{\text{tmpl}} \equiv [\sum_b (C_b^{EB, \text{th}} / \sigma_b^{\text{MC}})^2]^{1/2}$, where $C_b^{EB, \text{th}} = \sin(2\beta) \cos(2\beta) C_b^{EE}$ is the injected *EB* band-power template and σ_b^{MC} is the per-bin scatter across the 500 realizations — the matched-template significance of the injected signal against single-realization noise, the appropriate quantity for evaluating the deconvolution pipeline. It is *not* the significance of the recovered angle itself. The sky-fraction sweep artifact (`c1_fsky_sweep.json`) confirms the identification: its `snr_template_canonical_def` values 32.98 and 28.81 at $f_{\text{sky}} = 0.85$ and 0.65 are consistent with $20.32\sqrt{f_{\text{sky}}/0.32}$ (33.12 and 28.96; within 0.5%). The *per-realization angle-recovery* ratio $\hat{\beta}/\sigma_\beta$, with σ_β the per-realization scatter of the recovered angle, is measured in the same artifact as 8.1 ($\sigma_\beta = 0.029^\circ$, $f_{\text{sky}} = 0.85$) and 7.2 ($\sigma_\beta = 0.033^\circ$, $f_{\text{sky}} = 0.65$); σ_β was not recorded for the original canonical $f_{\text{sky}} = 0.32$ run; a dedicated 500-MC rerun at $f_{\text{sky}} = 0.32$ measures it directly as $\sigma_\beta = 0.046^\circ$ with $|\hat{\beta}|/\sigma_\beta = 5.2$ per realization (`reproducibility/p1_namaster_500mc/results/c9f_negative_beta.json`), confirming the $\sqrt{f_{\text{sky}}}$ -scaling estimate ($\approx 0.047^\circ$, ≈ 5). The pipeline-recovery SNR (template-fit significance of the injected signal) is 20.32 for the $\beta = 0.27^\circ$ injection. The same rerun injects $\beta = -0.27^\circ$ and recovers -0.238° (bias $+0.032^\circ$): recovery is sign-symmetric, with bias magnitude identical to the $+0.27^\circ$ injection. The standard error of the 500-MC mean is smaller by $\sqrt{N} = 22.4$.

⁴ The “pipeline-recovery SNR= 20.32” figure (and analogously

which matches the ACT pipeline footprint; the -60° lower edge approximates the ACT DR6 survey boundary in the sky-fraction sweep; apodized at 2° FWHM) galactic-cut masks with the same apodization recipe, noise level, and $N=500$ MC realizations ($N_{\text{side}} = 512$). The recovered values are $\hat{\beta} = 0.237^\circ$ ($f_{\text{sky}} = 0.85$, per-realization $\sigma_\beta = 0.029^\circ$) and $\hat{\beta} = 0.236^\circ$ ($f_{\text{sky}} = 0.65$, $\sigma_\beta = 0.033^\circ$), i.e. a recovery bias of -0.033° to -0.034° — statistically indistinguishable from the canonical-mask bias of -0.032° (the 0.001° and 0.002° differences are $0.8\times$ and $\approx 1.4\times$ the respective standard errors of the 500-realization means, $\sigma_\beta/\sqrt{500} \approx 0.0013^\circ$ and 0.0015°). The $\sim 12\%$ multiplicative under-recovery is therefore a sky-fraction-independent property of the pipeline, not an artifact of the small canonical mask.

Robustness battery and bias attribution.—A six-configuration robustness battery ($N = 500$ MC each, identical seeds to the canonical run; artifact `reproducibility/p1_namaster_500mc/results/c10_robustness_battery.json`) pins down the origin of the bias. First, an independent local rerun of the canonical configuration reproduces the pod anchor exactly ($\hat{\beta} = 0.238^\circ$, bias -0.032°). The bias is then *unchanged* under an apodization-scale sweep (0.5° and 3° FWHM vs. the canonical 2° : $\hat{\beta} = 0.239^\circ$ and 0.238°), under a larger Galactic cut ($|b| > 30^\circ$, $f_{\text{sky}} = 0.20$: 0.238°), and under B -mode purification (`purify_b=True`: 0.238°) — so the earlier attribution of the residual to apodization-induced power suppression is *not* supported by direct variation of the apodization scale, and the bias is likewise independent of footprint geometry and purification. Two mechanisms do move it: (i) replacing the unweighted χ^2 template fit with an inverse-variance-weighted fit recovers $\hat{\beta} = 0.264^\circ$ (bias -0.006°), removing $\approx 80\%$ of the bias — the unweighted fit’s equal weighting of noise-dominated high- ℓ bins is the dominant contribution; and (ii) replacing the crude $C_\ell^{BB} = 0.05 C_\ell^{EE}$ proxy with a CAMB lensed- Λ CDM BB spectrum recovers $\hat{\beta} = 0.251^\circ$ (bias -0.019°), consistent with the empirical $-C_\ell^{BB}$ template-mismatch robustness test described above (the ≈ 5 percentage-point reduction in bias is the empirical effect of injecting CAMB lensed- Λ CDM BB versus the proxy $0.05 C_\ell^{EE}$, not a closed-form analytic derivation). (Clarification of what this swap varies: per the committed battery script `reproducibility/p1_namaster_500mc/scripts/c10_robustness_battery.py`, the fit template is $\sin(2\beta)\cos(2\beta)C_\ell^{EE}$ with no BB term in *all* configurations; the `camb_lensed` swap changes the BB shape of the *injected synthetic skies* only, so the test isolates the dependence of the bias on the injected- BB shape against a fixed EE -only template, not a template-shape substitution.) A matched extension config (same 500 MC skies and seeds, CAMB lensed- Λ CDM BB injected and a $-C_\ell^{BB}$ -bearing template $\sin(2\beta)\cos(2\beta)(C_\ell^{EE} - C_\ell^{BB})$) recovers $\hat{\beta} = 0.251^\circ$ (bias -0.019°), unchanged from the EE -only-template configuration at the 10^{-3} -degree

fit-grid resolution — carrying the BB term in the template produces no further shift because the lensed BB is negligible against the synthetic EE template amplitude, so the residual -0.019° is not attributable to the remaining template-shape mismatch (config `camb_bb_template` in the same artifact). Restricting the fit to bins with $\ell \leq 1024$ changes nothing (0.238°): the noise-only bins above the band limit carry zero template weight. Because the validation carries the bias forward *empirically* as a multiplicative correction and a 0.040° observed pipeline bias (deconvolution-algebra bias on foreground-free skies; not a real-sky bias bound), none of the quoted headline numbers change; the battery sharpens only the attribution (estimator weighting + BB template shape, not apodization or masking). The total bias remains well below the published observational uncertainty ($\sigma_\beta^{\text{obs}} = 0.094^\circ$ from WMAP+Planck [5]; note $0.040^\circ \approx 0.5\sigma$ of the ACT DR6 $\sigma_\beta = 0.074^\circ$ [4], which is a published sky measurement and likewise not directly comparable to this synthetic-sky pipeline figure) at every sky fraction tested. *Canonical estimator choice.*—The unweighted χ^2 template fit is adopted as the canonical baseline to match the estimator configuration used in the public NaMaster driver scripts released by the published birefringence analyses (e.g., [5]), facilitating direct comparison of bias properties. The inverse-variance-weighted fit is evaluated in the robustness battery as a cross-check confirming that the dominant source of the $\sim 12\%$ multiplicative bias is the equal-weighting of noise-dominated high- ℓ bins; it is not adopted as the canonical estimator here precisely because that change would break direct comparability with the published estimator configuration. Artifacts: `reproducibility/p1_namaster_500mc/results/c10_robustness_battery.json`; `reproducibility/p1_namaster_500mc/results/c10_robustness_battery.json`.

V. COSMOLOGICAL FITS AND MODEL COMPARISON

A. Datasets and Configuration

We analyze four dataset combinations: (1) Planck CMB (NPIPE (PR4) CamSpec high- ℓ TTTEEE + 2018 low- ℓ TT/EE + 2018 lensing) [21] (the NPIPE/PR4 release provides the high- ℓ CamSpec likelihood; the low- ℓ and lensing likelihoods are taken from the 2018 release, the standard pairing in the Cobaya likelihood stack); (2) +SDSS BAO — eBOSS DR16 LRG/QSO/Ly α (auto and $\times$ QSO) [22] + SDSS DR7 MGS [23] + 6dFGS [24]; (3) +Pantheon+ [15]; (4) +SH0ES M_B -anchor [9] + DES Y3 S_8 [13]. No DESI BAO likelihood enters the frozen Λ CDM+ ΔN_{eff} chains; DESI DR2 BAO enters only the separate iter2 w_0w_a chain of Table II. The exact Cobaya likelihood blocks of all five chains are listed in Table III. For scope clarity: Table I summarizes only the two *frozen* Λ CDM+ ΔN_{eff} chains

(Planck+BAO+SN and full-tension), Table II reports the methodologically separate iter2 w_0w_a chain, and the Planck-only (accumulating) and Planck+BAO (diagnostic, not in headline) rows of Table III feed neither table. Parameter estimation uses Cobaya [25] (v3.5 original; v3.6.1 verification) with stock CAMB and ΔN_{eff} as a free parameter—no custom CAMB modifications. The extended parameter space adds ΔN_{eff} to ΛCDM , with both $(\omega/H)_0$ and Ω_k fixed to zero in the actual sampled YAML configuration (Sec. II). Reproducibility materials at <https://github.com/Hubify-Projects/bigbounce/tree/main/reproducibility>.

B. Results: $\Lambda\text{CDM}+\Delta N_{\text{eff}}$ proxy

The two load-bearing posteriors from the frozen chains are $\Delta N_{\text{eff}} = -0.020 \pm 0.169$ (full-tension) and $+0.058 \pm 0.179$ (Planck+BAO+SN), both consistent with zero. The $\Lambda\text{CDM}+\Delta N_{\text{eff}}$ proxy thus offers neither posterior preference nor exclusion at the present data precision. The ΔN_{eff} extension alone does not resolve the Hubble tension.

Independent re-run cross-check (this version).—A dedicated re-run of the Planck+BAO+SN $\Lambda\text{CDM}+\Delta N_{\text{eff}}$ configuration (matching likelihood stack for the SDSS BAO block and high- ℓ CamSpec component; the c15 low- ℓ EE and lensing likelihood names, `planck_2020_lollipop.lowlEE` and `planckpr4lensing`, differ from the frozen-chain names in Table III, `planck_2018_lowl.EE` and `planck_2018_lensing.clik`; the pairing mismatch is noted as a known limitation in the *Release-pairing note* below) converged on an independent compute pod to $R-1=0.0147$ (40,349 raw rows, 28,245 post-burn-in at 30%, 107,853 GetDist sum-of-weights — not the integrated-autocorrelation ESS) and gives $\Delta N_{\text{eff}} = +0.0514 \pm 0.171$, in 0.04σ agreement with the frozen $+0.058 \pm 0.179$ quote above. The auxiliary posteriors $H_0 = 67.81 \pm 1.07$ km/s/Mpc, $\sigma_8 = 0.813 \pm 0.009$, $S_8 = 0.828 \pm 0.010$, $\Omega_m = 0.311 \pm 0.006$ all reproduce the frozen-chain Table I values within $< 0.1\sigma$. The “consistent with zero / no Hubble-tension resolution” conclusion above is therefore reproduced on an independent chain; full chain artifacts and posterior summary are committed at [reproducibility/cosmology/chains/w0wa_quintom_desi_dr2/c15_converged/](https://reproducibility.cosmology/chains/w0wa_quintom_desi_dr2/c15_converged/). *Release-pairing note:* the primary frozen chains use Planck NPIPE (PR4) CamSpec high- ℓ TTTEEE paired with `planck_2018_lowl.EE` and `planck_2018_lensing.clik` (Planck 2018 releases). The c15 verification re-run instead uses `planck_2020_lollipop.lowlEE` and `planckpr4lensing` for the low- ℓ EE and lensing components, making c15 a release-pairing robustness rerun (PR4-consistent low- ℓ /lensing) rather than an identical-likelihood rerun of the frozen chains. The 0.04σ agreement in ΔN_{eff} across this likelihood substitution provides an empirical bound on pairing-induced bias at the quoted precision (see Sec. III, limitation note).

C. w_0w_a cross-check with stated SN-overlap systematic

The DESI DR2 w_0w_a chain (Table II) is reported as a caveated overlap-uncorrected diagnostic of the quintom-B direction on the existing converged Cobaya chain. Its SN likelihood combines DES-SN5YR and Pantheon+ without a joint overlap covariance (caveat (e) of Sec. III), so the SN-overlap systematic is stated explicitly as a known limitation of this product-likelihood combination and no model-selection statement is claimed. The goodness-of-fit decomposition (χ^2_{BAO} , χ^2_{CMB} , χ^2_{SN}) is reported in Table II; model-comparison statistics (AIC, BIC, $\ln B$) are *not* reported there because a robust $\ln B$ against ΛCDM requires dedicated nested sampling (the ΛCDM point is unsampled by the Metropolis-Hastings chain, lying deep in the joint marginal tails of the overlap-uncorrected posterior; see fn. a). The chain posterior readouts are $w_0 = -0.812 \pm 0.044$, $w_a = -0.667 \pm 0.186$, and $w_0 + w_a = -1.48 \pm 0.15$ (phantom-crossing region) **(no σ -distance or significance from ΛCDM is quoted: the uncorrected DES-SN5YR $\times$ Pantheon+ $\sim 20\%$ shared-event overlap double-counts the common supernovae and artificially narrows the posterior, so any departure-from- ΛCDM significance computed from these widths would be an overlap-inflated product-likelihood artifact, not a valid detection significance; this must be read together with caveat (e), fn. a); these readouts carry the overlap-uncorrected SN systematic of caveat (e) and are discussed in detail in Sec. III (Physics interpretation paragraph). The displacement of w_0 from -1 cannot be expressed as a significance: ΛCDM is unsampled in the present Metropolis-Hastings run, and the marginal posterior width that any such distance would use is artificially narrowed by the uncorrected SN overlap (fn. a); a valid exclusion bound requires the deferred overlap-controlled nested-sampling analysis. *Compatibility-only scope:* all results in this subsection are posterior compatibility diagnostics; all model-preference inference is deferred — no analysis here constitutes evidence for or against a bounce cosmology. *Relation to the bounce-class taxonomy of Paper I(a) [1]:* the quintom-B direction probed here is only one branch of the broader bouncing-cosmology landscape (minimal matter-bounce, ekpyrotic, Cuscuton, string-gas, and quintom variants), which carry distinct w_0w_a and f_{NL} signatures; this cross-check does not select among those branches and does not test the minimal matter-bounce prediction $f_{\text{NL}} = -35/8$, which is established structurally in Paper I(a).**

VI. COSMIC BIREFRINGENCE: SPECTATOR ALP CONSISTENCY CHECK

Note (spectator-status caveat, main text).—The headline $\beta \approx 0.27^\circ$ benchmark of this subsection requires a

TABLE III. Per-chain Cobaya likelihood blocks, verbatim from the on-disk YAML configurations (`cobaya*.yaml`; frozen-chain `spintorsion.input.yaml`; iter2 `spintorsion.dr2.input.yaml`). All four Λ CDM+ ΔN_{eff} chains share the Planck block `planck.NPIPE.high1.CamSpec.TTTEE + planck.2018.low1.TT + planck.2018.low1.EE + planck.2018.lensing.click`; the SDSS BAO block is `bao.sixdf.2011.bao + bao.sdss.dr7_mgs + bao.sdss.dr16.baoplus_{lrg,qso,lyauto,lyxqso}`. The first four rows are incremental: each adds to the stack of the row above. The iter2 chain uses `planck.2018.lensing.native` in place of `.click`.

Chain	Likelihood stack	Status
Planck-only	Planck block	accumulating (not frozen)
Planck+BAO	+ SDSS BAO block	diagnostic (not in headline)
Planck+BAO+SN	+ <code>sn.pantheonplus</code>	frozen (Table I)
Full-tension	+ <code>H0.riess2020Mb</code> + DES-Y3 S_8 Gaussian	frozen (Table I)
iter2 w_0w_a	<code>bao.desi.dr2.desi_bao.all</code> + Planck block + <code>sn.desy5</code> + <code>sn.pantheonplus</code>	converged (Table II)

$\gtrsim 100\times$ fine-tuning of the misalignment initial condition under a $\cos\theta_i$ -flat prior (equivalently $\sim 25\times$ relative to the ad-hoc $\theta_i \approx 0.5$ midpoint; $\theta_i \sim 0.1$ versus the natural-prior midpoint $\theta_i \sim 0.5$; quantitative derivation in fn. 6) to keep the ALP an actual spectator field; this caveat is load-bearing for the “consistency-not-prediction” framing and is stated here, in the main text, rather than in a footnote only. The operational criterion for spectator status throughout this section is $\Omega_a < 0.01$: samples satisfying this cut constitute 13% of the full ALP posterior mass (Table IV) and are the only subset for which the ALP dark-energy fraction is safely sub-dominant. The threshold value 0.01 is physically motivated rather than tuned to a target: $\Omega_a = \rho_a(z=0)/\rho_{\text{crit},0}$ is the present-day ALP fraction of the critical density (defined in §VI), so $\Omega_a < 0.01$ requires the field to carry less than 1% of the energy budget today — a conservative spectator bound one full decade stricter than the nominal sub-dominance scale $\Omega_a < 0.1$. We deliberately quote the spectator-safe verdict at this stricter end. The cut-dependence is reported explicitly, not hidden: the companion subset table (Table IV) lists both the $\Omega_a < 0.1$ (44% mass, ESS= 1989) and $\Omega_a < 0.01$ (13% mass, ESS= 461) readouts, and the rotation marginal β is stable across them (the verdict does not flip between the two thresholds); we do not attempt a Bayes-factor cut, which would require a model-comparison the present overlap-uncorrected stack cannot support. This subsection presents a secondary consistency check using the spectator ALP model. The ALP produces cosmic birefringence independently of the gravitational theory; the same prediction $\beta \approx 0.27^\circ$ arises in standard GR with an identical ALP Lagrangian within the scan-prior envelope but near its upper-displacement/coupling edge; the posterior-supported fixed- $C_{a\gamma} = 8$ fit shifts to $m \gg H_0$ (median $\simeq 36 H_0$), with the same $\sim 25\times$ misalignment tuning required to recover the birefringence signal disclosed below and in fn. 6. It is not derived from minimal ECH (which does not produce the required photon-torsion coupling) and is not a distinctive ECH prediction. The model class was previously studied by Fujita *et al.* [26].

Headline observational constraint.—The primary observational reference adopted in this analysis is the pub-

lished Eskilt & Komatsu joint WMAP+Planck value $\beta = 0.342^\circ \pm 0.094^\circ$ (3.6σ) [5] (the joint WMAP9 + Planck PR3 analysis as published; the PR4/NPIPE label refers to the repository rerun discussed in the disambiguation footnote earlier in this paper; ACT DR6 enters only via the separate $\beta = 0.215^\circ \pm 0.074^\circ$ measurement [4], used below only in the auxiliary inverse-variance combination), which accounts for shared calibration systematics. The simplified inverse-variance combination below (3.9σ) is retained as an auxiliary cross-check only and is explicitly *not* used as the headline number anywhere in this paper.

ALP field evolution.—Numerical integration of the ALP equation of motion $\ddot{\phi} + 3H\dot{\phi} + m^2 f_a \sin(\phi/f_a) = 0$ in a Λ CDM background⁵ yields the field displacement from recombination to today, at the representative corner $m = 2H_0$, $\theta_i = 1$ of the natural-parameter box:

$$\Delta\phi/f_a \approx 0.42 \quad (m = 2H_0, \theta_i = 1). \quad (3)$$

Across the natural parameter range $m/H_0 \in [1, 3]$, $\theta_i \in [0.5, 2]$ the committed EOM grid gives $\Delta\phi/f_a \in [0.064, 1.19]$ (`research/branch_R_alp_birefringence/phase2_mcmc/c10b_alp_envelope_scan.json`).⁶

⁵ The ALP ODE is integrated on a Λ CDM late-time $H(z)$ here as an empirical background, distinct from the quintom-bounce dynamics that supply the early-universe / contracting-phase $H(z)$ in the underlying ECH cosmology [27]. The Λ CDM-background choice is conservative for the ALP-MCMC: a quintom late-time w_0w_a background (as in Sec. V) shifts $H(z)$ at $z \lesssim 1$ by $\sim$ few percent, propagating to a $\lesssim$ few-percent systematic on $\Delta\phi/f_a$ — well below the $\sim 30\%$ prior-width envelope on θ_i and m/H_0 dominating the β prediction.

⁶ Backreaction disclosure: the ALP backreaction fraction scales as $\Omega_a \sim \rho_a/\rho_{\text{crit}} \sim (m^2 f_a^2 / H_0^2 M_{\text{Pl}}^2) \theta_i^2$, so the abstract’s spectator-status restriction $\theta_i \ll 1$ (see the *Spectator-status caveat* in the abstract) implies $\Omega_a \ll 1$ only in the sub-natural sliver $\theta_i \sim 0.1$. The numerical scan range $\theta_i \in [0.5, 2]$ is RETAINED here for completeness of the parameter envelope, but the natural-prior-anchored spectator-consistent result sits at $\theta_i \sim 0.1$: at $\theta_i = 0.1$ vs the scan-midpoint $\theta_i = 0.5$ the backreaction is $\Omega_a(0.1)/\Omega_a(0.5) \sim 1/25$ (i.e., a $\sim 25\times$ fine-tuning of the misalignment initial condition is required to keep the ALP a true spectator). The $\Omega_a \sim 1$ regime at $\theta_i \sim 1$ is the dark-energy-ALP regime explicitly excluded from this spectator-consistency companion check.

Birefringence value.—For $C_{a\gamma} = 8$, $\theta_i = 1$, $m \approx 3.9 H_0$ (the committed EOM integration gives $\Delta\phi/f_a = 1.06$ there; artifact `research/branch_R_alp_birefringence/phase2_mcmc/c10b_alp_envelope_scan.json`; an independent fixed-step fourth-order Runge–Kutta re-integration of the same EOM in $\ln a$ with the frozen initial condition $\dot{\theta} = 0$ at $z_{\text{init}} = 3000$ reproduces $\Delta\phi/f_a = 1.0601$ to four significant figures):

$$\begin{aligned} \beta &\approx \frac{\alpha_{\text{EM}}}{4\pi} C_{a\gamma} \frac{\Delta\phi}{f_a} \\ &= \underbrace{(7.297 \times 10^{-3}/(4\pi))}_{\alpha_{\text{EM}}/(4\pi) \approx 5.81 \times 10^{-4}} \times 8 \times 1.06 \\ &= 4.93 \times 10^{-3} \text{ rad} \times \frac{180^\circ}{\pi} \approx 0.28^\circ. \end{aligned} \quad (4)$$

The product of the three factors is $5.81 \times 10^{-4} \times 8 \times 1.06 = 4.93 \times 10^{-3} \text{ rad}$ (with $\alpha_{\text{EM}} = 1/137.036$; rounding to two significant figures on $\Delta\phi/f_a$ would give 4.65×10^{-3} , so the third significant figure is sensitive to the precision of the EOM integration — here we use the four-figure $\Delta\phi/f_a = 1.0601$ from the committed Runge–Kutta re-integration). The $\alpha_{\text{EM}}/(4\pi)$ prefactor is convention-dependent ($\alpha_{\text{EM}}/(2\pi)$ appears in some Lagrangian normalizations); here it corresponds to the normalization $\mathcal{L} \supset -(g_{a\gamma}/4) \phi F_{\mu\nu} \tilde{F}^{\mu\nu}$ with $g_{a\gamma} = C_{a\gamma} \alpha_{\text{EM}}/(2\pi f_a)$ and $\beta = (g_{a\gamma}/2) \Delta\phi$, the convention fixed in the committed model specification and implemented identically throughout the numerical pipeline (`research/branch_R_alp_birefringence/02_birefringence_prediction.md`; `research/branch_R_alp_birefringence/phase2_mcmc/alp_ode.py`). The fiducial value $\beta \approx 0.27^\circ$ corresponds at $C_{a\gamma} = 8$ to EOM trajectories with, e.g., $(\theta_i = 1, m \approx 3.9 H_0)$ or $(\theta_i \approx 1.4, m \approx 3 H_0)$. The committed EOM integration (`research/branch_R_alp_birefringence/phase2_mcmc/alp_ode.py`) gives $\Delta\phi/f_a = 0.35\text{--}0.42$ at masses $m \approx 1.8\text{--}2 H_0$ ($\theta_i = 1$); all mass pairings are verified against the released grid scan. Across the box $C_{a\gamma} \in [4, 12]$, $m/H_0 \in [1, 3]$, $\theta_i \in [0.5, 2]$ the committed EOM gives $\beta \approx 0.01\text{--}0.48^\circ$ (grid scan over physical trajectories; same artifact; the span is the *union* over the full $C_{a\gamma} \times (m/H_0, \theta_i)$ box — the lower edge is reached at $C_{a\gamma} = 4$ and the upper at $C_{a\gamma} = 12$, not at any fixed coupling), a span that brackets the observed value but reaches it only in the upper-right corner of the box ($\beta(C_{a\gamma}=8)$ peaks at 0.32° at $\theta_i = 2, m = 3 H_0$; the observed 0.342° at $C_{a\gamma} = 8$ requires $m \gtrsim 4 H_0$ or larger coupling). See Appendix C for the full ALP-MCMC sampled-parameter list, priors, and dataset-likelihood configuration. The spectator-consistent corner of this envelope ($\theta_i \sim 0.1$, per fn. 6) does require a $\sim 25\times$ tuning of the misalignment initial condition relative to the natural prior midpoint $\theta_i \sim 0.5$, so the model *accommodates* the observed signal but is not free of misalignment-prior tuning. The $[0.01, 0.48]^\circ$ span quoted above is computed point-by-point on physical EOM trajectories ($\Delta\phi/f_a$ is a

derived function of $(m/H_0, \theta_i)$, not an independent variable); the committed grid scan gives $\beta_{\text{ALP}} \in [0.01, 0.48]^\circ$ across the full scan-prior box (the benchmark EOM grid, not the spectator-consistent subset).

Summary-likelihood combination (auxiliary cross-check).—Combining $\beta = 0.30^\circ \pm 0.11^\circ$ (Planck NPIPE [3]) and $\beta = 0.215^\circ \pm 0.074^\circ$ (ACT DR6 [4]) via inverse-variance weighting:

$$\beta_{\text{combined}} = 0.241^\circ \pm 0.061^\circ \quad (\text{naive } 3.9\sigma \text{ upper bound}) \quad (5)$$

Note: this 3.9σ is a deliberately-optimistic upper bound assuming zero Planck–ACT correlation; see Eskilt & Komatsu 3.6σ for the properly-correlated headline. (Auxiliary cross-check only; the 3.9σ is the significance of β_{combined} relative to the null $\beta = 0$, i.e. $0.241/0.061$, under the deliberately-optimistic zero-correlation assumption only.) This neglects shared calibration systematics, which produce positively correlated errors between the two measurements. Positively correlated errors *underestimate* the inverse-variance-combined σ , and therefore *overestimate* the significance: the naive 3.9σ figure is an upper bound on the true significance, not a lower bound. The published joint analysis at 3.6σ [5], which properly accounts for the shared polarization-angle calibration systematics via a joint covariance matrix with the Tau A self-calibration nuisance parameters, is the headline; the 3.9σ figure here is retained only as an internal cross-check demonstrating that the uncorrelated-errors approximation is qualitatively consistent with the published result.

MCMC parameter estimation.—Dedicated MCMC sampling of the ALP parameter space (three committed configurations totaling 9,720 accepted samples: $C_{a\gamma} = 8$ fixed with θ_i and $\log_{10} m_a$ sampled, 2,160; $C_{a\gamma} \in [1, 30]$ additionally sampled, 6,840; model-independent β_{free} , 720; chain configurations and per-chain counts in Appendix C) yields: $\beta_{\text{ALP}} = 0.336^\circ \pm 0.10^\circ$ ($C_{a\gamma} = 8$ fixed; the direct-sample priors are on the underlying ALP parameters (θ_i, m_a) , not on $\Delta\phi/f_a$, which is a derived quantity along each ALP trajectory. The MCMC posterior, anchored to the Gaussian summary likelihood on the published Eskilt–Komatsu joint WMAP+Planck measurement (fn. 3; not to the *EB* spectra themselves; the uniform-rotation observable is periodic, $\beta \equiv \beta + n \times 90^\circ$ for E/B , and the Gaussian summary ignores wrapping — harmless here because the posterior support is confined to $|\beta| \lesssim 0.7^\circ \ll 90^\circ$, so wrapped images carry negligible likelihood), settles at $\theta_i = 1.32 \pm 0.41$ and $m \sim 10\text{--}10^2 H_0$ (posterior median $m \approx 36 H_0$) — i.e. *outside* the natural envelope box $\theta_i \in [0.5, 2]$, $m/H_0 \in [1, 3]$ in mass — where the data-preferred joint product is $C_{a\gamma}(\Delta\phi/f_a) \approx 10.3$, corresponding at the fixed $C_{a\gamma} = 8$ to $\Delta\phi/f_a \approx 1.29$, $\sim 8\%$ above the box maximum $\Delta\phi/f_a = 1.19$ ($\theta_i = 2, m = 3 H_0$; grid-scan artifact above); this is the same fine-tuning-prone accommodation regime flagged in fn. 6 and is consistent with the model accommodating but not naturally explaining the observed $\beta_{\text{obs}} = 0.342^\circ$), consis-

tent with the model-independent fit $\beta_{\text{free}} = 0.344^\circ \pm 0.10^\circ$ (our internal model-independent MCMC fit, with β as a free parameter, to a Gaussian summary likelihood on the published ESKILT-KOMATSU joint WMAP+Planck birefringence measurement $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$ [5] — not a re-analysis of the EB spectra themselves; 720 accepted samples in the dedicated β_{free} configuration; full priors and likelihood details in Appendix C; β_{free} denotes the unconstrained-amplitude fit distinct from β_{ALP} which has $C_{a\gamma} = 8$ fixed) and the observed $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$. All three within 1σ (e.g. β_{ALP} vs β_{obs} : the model posterior mean 0.336° lies 0.06σ from the data central value, $\Delta = 0.006^\circ$, using the data uncertainty $\sigma_{\text{obs}} = 0.094^\circ$; these are consistency statements against the same single published measurement, not independent confirmations). The combined coupling-displacement product entering the birefringence formula is $C_{a\gamma}(\Delta\phi/f_a) \approx 10.3$ ($\beta = 0.342^\circ$ in radians is 5.97×10^{-3} , the prefactor $\alpha_{\text{EM}}/(4\pi)$ is 5.8×10^{-4} , giving $C_{a\gamma}\Delta\phi/f_a = \beta/[\alpha_{\text{EM}}/(4\pi)] \approx 10.3$); with the field-displacement range $\Delta\phi/f_a \in [0.064, 1.19]$ from the committed EOM grid over the natural box, the required $C_{a\gamma}$ spans ≈ 8.6 (largest displacement: $\theta_i = 2$, $m = 3H_0$) up to ≈ 160 (smallest displacement: $\theta_i = 0.5$, $m = H_0$; note the continuous-prior scan below covers $C_{a\gamma} \in [4, 60]$ only — small-displacement corners with $\Delta\phi/f_a \lesssim 0.17$ requiring $C_{a\gamma} > 60$ lie outside that scan and carry negligible posterior support); at the posterior-preferred masses ($m \sim 10\text{--}10^2 H_0$, where the displacement saturates near $\Delta\phi/f_a \approx 1.2\text{--}1.3$), the required coupling is $\approx 8\text{--}10$. Even the lower end exceeds the standard KSVZ/DFSZ benchmark range, which predicts $|C_{a\gamma}| \sim \mathcal{O}(1)$; the entire required range therefore lies outside minimal ALP photon-coupling benchmarks and requires non-minimal model building. We emphasize that this coupling burden and the misalignment burden are *two independent* fine-tunings: a non-minimal photon coupling ($C_{a\gamma} \gtrsim 9$) is required regardless of θ_i , while the spectator-consistency restriction additionally fine-tunes the misalignment initial condition to $\theta_i \sim 0.1$ (fn. 6); neither tuning substitutes for the other. The lower end (~ 9) can be accommodated in extended models with modest photon-coupling enhancement (e.g. chiral-fermion-loop enhancement, clockwork constructions), while light-mass / small-displacement corners of the box demand couplings ($\gtrsim 50\text{--}160$) requiring substantial UV-completion enhancement. The signal is therefore accommodated across the considered parameter space rather than fine-tuned only at one benchmark, but the upper-coupling end is not generic. We emphasize that the spectator-consistent corner $\theta_i \sim 0.1$ (fn. 6) inflates the required photon-coupling further: at fixed $\beta = 0.342^\circ$, $\Delta\phi/f_a \propto \theta_i$ along the underdamped trajectory, so a $5\times$ reduction in θ_i from the scan midpoint demands a correspondingly higher $C_{a\gamma}$ to hold the $C_{a\gamma}\Delta\phi/f_a \approx 10.3$ product, pushing the required enhancement well above standard KSVZ/DFSZ $\mathcal{O}(1)$ benchmarks. Because the required coupling extends well

beyond the $C_{a\gamma} \in [1, 30]$ prior of the original extended configuration (which truncated $\sim 28\%$ of the posterior mass above $C_{a\gamma} = 30$; the truncation fraction is computed from the `run2_extended` chain of Appendix C), we reran the $C_{a\gamma}$ -free fit with a continuous uniform prior $C_{a\gamma} \in [4, 60]$ (4 MPI chains, $\hat{R} - 1 = 0.0095$, 8,955 accepted samples); this prior covers the required coupling for every trajectory with $\Delta\phi/f_a \gtrsim 0.17$, including the entire posterior-preferred saturated-displacement regime, while the small-displacement corners requiring $C_{a\gamma} > 60$ carry negligible joint-posterior support (only 5% of the posterior mass sits above $C_{a\gamma} = 55$, with no pile-up at the prior edge). The resulting posterior is broad, as expected for a single-amplitude constraint on a three-parameter degeneracy: median $C_{a\gamma} = 20.7$ with 16–84% range [7.3, 45.6]; 69% of the posterior mass falls inside [9, 51], and 22% lies below $C_{a\gamma} = 9$, consistent with the high- $\Delta\phi/f_a$ tail, where the fixed product is met at smaller coupling. These mass fractions are prior-dependent statements (flat $C_{a\gamma} \in [4, 60]$, $\theta_i \in [0.01, \pi]$, $\log_{10} m_a$ priors), not prior-independent measurements. In particular the θ_i prior is flat in the angle; the vacuum-manifold-uniform alternative (flat in $\cos\theta_i$, density $\propto \sin\theta_i$) carries *less* mass at small θ_i , so the quoted spectator-sliver posterior fractions would decrease further under that prior swap — the flat- θ_i choice is the more generous one for the spectator corner, and the accommodation-not-natural conclusion is robust in direction to this prior choice. A direct rerun under the $\cos\theta_i$ -flat prior (density $\propto \sin\theta_i$; 10,800 accepted samples, $\hat{R} - 1 = 0.008$) confirms this empirically: the coupling posterior is essentially unchanged (median $C_{a\gamma} = 17.1$, 16–84% [6.8, 43.4]; $\beta = 0.328^\circ \pm 0.100^\circ$) while the $\theta_i \leq 0.1$ spectator-sliver mass drops from 0.33% to 0.068% (`research/branch_R_alp_birefringence/phase2_mcmc/chains/c14_costheta/c14_summary.json`). The recovered $\beta = 0.326^\circ \pm 0.099^\circ$ posterior matches the observed $0.342^\circ \pm 0.094^\circ$, confirming the consistency-check verdict over the posterior-supported coupling range rather than at fixed benchmarks. *Spectator-subset read-out* (same chain, no additional sampling): the chain’s derived ALP energy fraction Ω_a (defined in the subsection below; computed under the small-angle quadratic approximation, valid for $\theta_i \ll 1$; the leading anharmonic correction is $\mathcal{O}(\theta_i^2/12)$, i.e. $\lesssim 8\%$ at $\theta_i \sim 1$ and $\lesssim 1\%$ over the posterior-supported $\Omega_a \leq 0.01$ spectator-safe subset ($\theta_i \lesssim 0.3$; Table IV), sub-dominant to the prior width) satisfies $\Omega_a < 0.1$ for 44% and $\Omega_a < 0.01$ for 13% of the posterior mass; restricted to the $\Omega_a \leq 0.01$ spectator-safe subset, the rotation marginal is $\beta = 0.28^\circ \pm 0.10^\circ$ (this is the subset posterior median; the injection fiducial 0.27° is the rounded value of the canonical injection point, distinct from this posterior marginal), consistent with $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$ at 0.5σ . *H_0 marginalization note:* Ω_a at each MCMC step uses the Λ CDM background Hubble rate evaluated at the Cobaya posterior mean $H_0 = 67.68 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (fixed, not marginalized).

Marginalizing H_0 over the Planck 1σ interval shifts Ω_a by $\lesssim 3\%$ ($\Omega_a \propto H_0^{-2}$), well below the statistical uncertainty of the $\Omega_a < 0.01$ cut. The strict $\theta_i \leq 0.1$ sliver of fn. 6 carries only 0.33% of the posterior mass by MC weight (42 of the 8,955 raw samples, i.e. 0.47% by raw count — the weighted fraction is lower because the sliver samples carry below-average MC weights — too few for a stable marginal, so the following band is indicative only): within that sliver the required coupling piles against the upper prior edge, with weighted $C_{a\gamma}$ 16/50/84 percentiles of 36.8/47.2/55.6 (flat prior $C_{a\gamma} \in [4, 60]$), well above the KSVZ/DFSZ $\mathcal{O}(1)$ benchmarks (research/branch_R_alp_birefringence/phase2_mcmc/c10a_spectator_slice.json). This quantifies the misalignment tuning of the spectator-consistent corner. Convergence: $R - 1 < 0.01$ for all runs. The joint posterior structure of this continuous-prior configuration is shown in Fig. 4. Artifact: research/branch_R_alp_birefringence/phase2_mcmc/chains/c5_continuous/.

ALP dark-energy fraction Ω_a : definition and computation

The spectator-status classification in this section uses the ALP dark-energy fraction $\Omega_a = \rho_a(z=0)/\rho_{\text{crit},0}$. We define this quantity precisely here because the Table IV entries (44% at $\Omega_a < 0.1$; 13% at $\Omega_a < 0.01$) depend on it directly, and two independent reviewers flagged the absence of an explicit derivation.

Axion potential.—The spectator ALP carries the standard cosine potential

$$V(\phi) = m_a^2 f_a^2 \left[1 - \cos\left(\frac{\phi}{f_a}\right) \right], \quad (6)$$

where m_a is the ALP mass, f_a the decay constant, and $\phi_i = \theta_i f_a$ the initial field value parameterized by the misalignment angle $\theta_i \equiv \phi_i/f_a$. For $\theta_i \ll 1$ the potential is approximately quadratic, $V \approx \frac{1}{2} m_a^2 \phi_i^2$; for $\theta_i \sim \mathcal{O}(1)$ anharmonic corrections enter at $\mathcal{O}(\theta_i^2/12)$.

Onset of oscillations.—The ALP is frozen by Hubble friction and begins coherent oscillations when

$$3H(z_{\text{osc}}) = m_a, \quad (7)$$

where $H(z)$ is evaluated on the Λ CDM background (with $H_0 = 67.68 \text{ km s}^{-1} \text{ Mpc}^{-1}$, fixed at the Cobaya posterior mean; see the H_0 marginalization note in the spectator-subset readout above). For ALP masses in the scan prior $m/H_0 \in [7 \times 10^{-3}, 7 \times 10^2]$, the onset redshift ranges from $z_{\text{osc}} \lesssim 0$ (lightest masses, ALP still frozen; behaves as a cosmological-constant contribution during the survey epoch) to $z_{\text{osc}} \gg 1$ (heaviest masses, early dark matter). Equation (7) is solved numerically at each MCMC step in research/branch_R_alp_birefringence/phase2_mcmc/alp_ode.py.

Energy density today.—Once oscillating ($z \ll z_{\text{osc}}$), the ALP cycle-averaged energy density redshifts as matter:

$$\rho_a(z) = \rho_a(z_{\text{osc}}) \left(\frac{1+z}{1+z_{\text{osc}}} \right)^3, \quad z \ll z_{\text{osc}}. \quad (8)$$

At onset the potential dominates: $\rho_a(z_{\text{osc}}) \approx V(\phi_i) = m_a^2 f_a^2 [1 - \cos(\theta_i)]$. The dark-energy fraction today is therefore

$$\Omega_a \equiv \frac{\rho_a(z=0)}{\rho_{\text{crit},0}} \approx \frac{m_a^2 f_a^2 [1 - \cos(\theta_i)]}{\rho_{\text{crit},0} (1+z_{\text{osc}})^3}, \quad (9)$$

with $\rho_{\text{crit},0} = 3H_0^2 M_{\text{Pl}}^2 \approx 3.7 \times 10^{-11} \text{ eV}^4$ and $M_{\text{Pl}} \approx 2.44 \times 10^{18} \text{ GeV}$. For $f_a = M_{\text{Pl}}$ and small θ_i this gives $\Omega_a \approx m_a^2 \theta_i^2 / (6H_0^2 (1+z_{\text{osc}})^3)$, confirming the $\Omega_a \propto \theta_i^2$ scaling of fn. 6.

Computation from sampled parameters.—At each MCMC step the ALP module receives (m_a, θ_i) (with $f_a = M_{\text{Pl}}$ fixed), solves Eq. (7) for z_{osc} , and evaluates Eq. (9) using the potential-dominated approximation, verified against full EOM integration for committed chain parameters. For the lightest masses with $z_{\text{osc}} \leq 0$ the ALP remains Hubble-frozen through the survey epoch and is held at $\rho_a = V(\theta_i)$ (a cosmological-constant-like contribution, as noted in the onset discussion above) rather than diluted by the $(1+z_{\text{osc}})^{-3}$ factor of Eq. (9). The per-step Ω_a values are stored in research/branch_R_alp_birefringence/phase2_mcmc/chains/c5_continuous/ and define the spectator-status cuts in Table IV below. The 44%/13% posterior-mass fractions are direct weighted integrals over these per-step values; they are prior-dependent (flat $\theta_i \in [0.01, \pi]$) and not prior-independent measurements of Ω_a .

LiteBIRD forecast.—LiteBIRD is projected to achieve $\sigma(\beta) \approx 0.03^\circ$ [28] (a forecast under foreground- and calibration-control assumptions, not a guaranteed instrument performance). For $\beta = 0.27^\circ$ this corresponds to $\sim 9\sigma$ statistical significance against the $\beta = 0$ null. This 9σ figure is *not* a model-discrimination forecast: separating the spectator-ALP value 0.27° from the current WMAP+Planck central value $0.342^\circ \pm 0.094^\circ$ is limited by the published measurement's uncertainty, $|0.342 - 0.27|/\sqrt{0.03^2 + 0.094^2} \approx 0.7\sigma$, so LiteBIRD alone will not separate the spectator-ALP value from the observed central value (the two figures correspond to distinct null hypotheses).

Caveats.—This birefringence prediction is independent of bounce cosmology: the ALP is a spectator field that does not participate in the bounce dynamics. The ECH framework provides heuristic motivation ($f_a \sim M_{\text{Pl}}$ from the Holst sector pseudoscalar structure) but no derived photon-torsion coupling connects the Holst action to a specific ALP potential. The model can *accommodate* the observed β within the scan-prior envelope but near its upper-displacement/coupling edge; the posterior-supported fixed- $C_{a\gamma} = 8$ fit shifts to $m \gg H_0$ (median

TABLE IV. Restricted-posterior readout of the continuous-prior `c5_continuous` chain on three spectator-status subsets, together with the full chain. Posterior-mass columns give the fraction of MC weight (and, for the strict $\theta_i \leq 0.1$ sliver, raw-count fraction in parentheses); β medians are direct chain readouts; m/H_0 , θ_i , and $C_{a\gamma}$ entries report medians or representative weighted statistics; ESS is the weight-expanded Sokal effective-sample size for the marker parameter on the indicated subset. *The fixed-coupling ($C_{a\gamma} = 8$) posterior-supported median sits at $m \simeq 36 H_0$ — well above the scan-prior $m \sim H_0$ benchmark — and only the $\Omega_a \leq 0.01$ spectator-safe subset enforces a true spectator while still consistent with the observed $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$ at $\sim 0.5\sigma$. The spectator-consistent verdict rests on this $\Omega_a < 0.01$ subset (13% mass, stable ESS = 461), not on the strict $\theta_i \leq 0.1$ sliver (0.33% weighted mass, 42 samples), which is reported as “indicative only.”*

Subset	post. mass	β (deg)	m/H_0	θ_i	$C_{a\gamma}$	ESS
full chain	100%	0.326 ± 0.099	median $\simeq 36$	broad	med. 20.7, [7.3, 45.6]	β 2860; θ_i 796
$\Omega_a < 0.1$	44%	0.328 ± 0.100	4.7/37.7/264	0.22/0.41/0.70	14.2/26.2/46.4	1989
$\Omega_a < 0.01$ (safe)	13%	0.28 ± 0.10	6.0/40.5/238	0.15/0.21/0.27	29.9/43.3/54.1	461
$\theta_i \leq 0.1$ (strict)	0.33% (0.47% raw)	indicative only	$\sim \mathcal{O}(1)$	≤ 0.1 (tuned)	16/50/84 = 36.8/47.2/55.6	sliver-only (42 samples)

$\simeq 36 H_0$). The spectator-safe interpretation requires a tuned misalignment subspace (the $\sim 25\times$ misalignment tuning required for the birefringence signal is disclosed in Sec. VI and fn. 6) and a non-minimal photon coupling; it does not uniquely predict the signal.

VII. CONCLUSIONS

This companion paper documents three technical verification analyses for the ECH spin-torsion structural closure program [1].

$\Lambda\text{CDM} + \Delta N_{\text{eff}}$ MCMC proxy.—The stock-CAMB Cobaya v3.6.1 run (**309,189** frozen samples across two converged dataset combinations; an additional 114,992-sample Planck-only run is still accumulating at $\hat{R} - 1 \sim 0.05$, is *not* reported in Table I, and is *not* aggregated into the frozen headline) finds ΔN_{eff} consistent with zero in both frozen dataset combinations and recovers $H_0 = 67.68 \pm 1.06 \text{ km s}^{-1} \text{ Mpc}^{-1}$, consistent with standard Planck ΛCDM . The ΔN_{eff} extension does not resolve the Hubble tension. We note that the ΔN_{eff} constraint reported here uses Planck PR4 / NPIPE CamSpec high- ℓ TTTEEE paired with Planck 2018 low- ℓ TT/EE + lensing (`planck.2018.lowl.EE + planck.2018.lensing.clik`); the c15 verification re-run uses the PR4-consistent counterparts `planck.2020.lollipop.lowlE` and `planckpr4lensing`, making it a release-pairing robustness rerun; the resulting 0.04σ agreement in ΔN_{eff} provides an empirical bound on pairing-induced bias at the quoted precision (see Sec. V, *Release-pairing note*). Current data neither require nor exclude a phenomenological ΔN_{eff} proxy contribution for the bounce class; CMB-S4 ($\sigma(N_{\text{eff}}) \sim 0.03$ [17]) will sharpen this constraint substantially. Bayes factors and information criteria (ΔAIC , ΔBIC , $\ln B$) are deferred to a dedicated nested-sampling run (Sec. V, *Model-comparison statistics* paragraph); we report only the parameter posteriors (ΔN_{eff} , H_0 , etc.) as load-bearing here and explicitly omit Bayes-factor or information-criterion comparisons in this Metropolis-Hastings-only analysis.

NaMaster pipeline validation.—The 500-MC pseudo- C_ℓ analysis on synthetic ΛCDM polarization skies con-

firms that the deconvolution pipeline recovers injected birefringence angles with amplitude-dependent bias $\hat{\beta} - \beta_{\text{inj}} = -0.032$ to -0.040° (worst-case -0.040° at injection $\beta = 0.342^\circ$; Sec. IV) at template-fit SNR consistent with the ACT-noise floor. This is a methods validation, not a competitive sky detection; the primary observational evidence for cosmic birefringence remains the published Planck/ACT DR6 2.7–2.9 σ measurements.

Spectator-ALP consistency.—To state the verdict first: the spectator ALP *accommodates*, it does not *predict*, the observed birefringence, and the spectator-safe verdict rests on the broader $\Omega_a < 0.01$ subset (13% of the posterior mass), not on the extreme $\theta_i \leq 0.1$ sliver (0.33% mass, indicative only). Within that $\Omega_a < 0.01$ spectator-safe subset (13% of the posterior mass; median $m \simeq 40.5 H_0$, $\Omega_a < 0.01$, ESS = 461), an ALP with $f_a \sim M_{\text{Pl}}$ and posterior median $m \simeq 40.5 H_0$ (not $m \sim H_0$; the scan-prior envelope spans $m \sim H_0$ but the posterior-supported fixed- $C_{a\gamma} = 8$ fit shifts well above the prior midpoint) is consistent with the published 3.6 σ joint signal, with the caveat that this spectator-safe subset ($\theta_i \sim 0.1$, a $\sim 25\times$ fine-tuning relative to the natural midpoint; fn. 6) satisfies spectator status ($C_{a\gamma} \Delta\phi/f_a \approx 10.3$ for $\beta = 0.342^\circ$, with $\Delta\phi/f_a \in [0.064, 1.19]$ over the natural box giving required $C_{a\gamma}$ from ≈ 8.6 up to ≈ 160 at the smallest displacements, and ≈ 8 –10 in the posterior-preferred saturated-displacement regime; see §VI for the explicit numerical derivation). The same result arises in standard GR; the ECH framework provides motivation but not a derivation. Under forecast foreground-cleaning and calibration assumptions [28], a $\beta \simeq 0.27^\circ$ signal would be detected relative to $\beta = 0$ at $\sim 9\sigma$ by LiteBIRD in the early 2030s.

$w_0 w_a$ cross-check.—The converged DESI DR2 + Planck NPIPE + Pantheon+ [15] + DES-SN5YR [14] Cobaya chain with the $w_0 w_a$ free-parameter extension (128,385 accepted samples across 16 MPI chains; $\hat{R} - 1 = 0.00820$, below the standard $\hat{R} - 1 < 10^{-2}$ publication target) supplies the GetDist $w_0 w_a$ posteriors of Table II, a caveated overlap-uncorrected diagnostic of the quintom-B scenario [27] (tested against DESI DR2 [19]) under an overlap-uncorrected SN product likelihood (Sec. III, caveat (e)); the SN-overlap systematic is stated as a

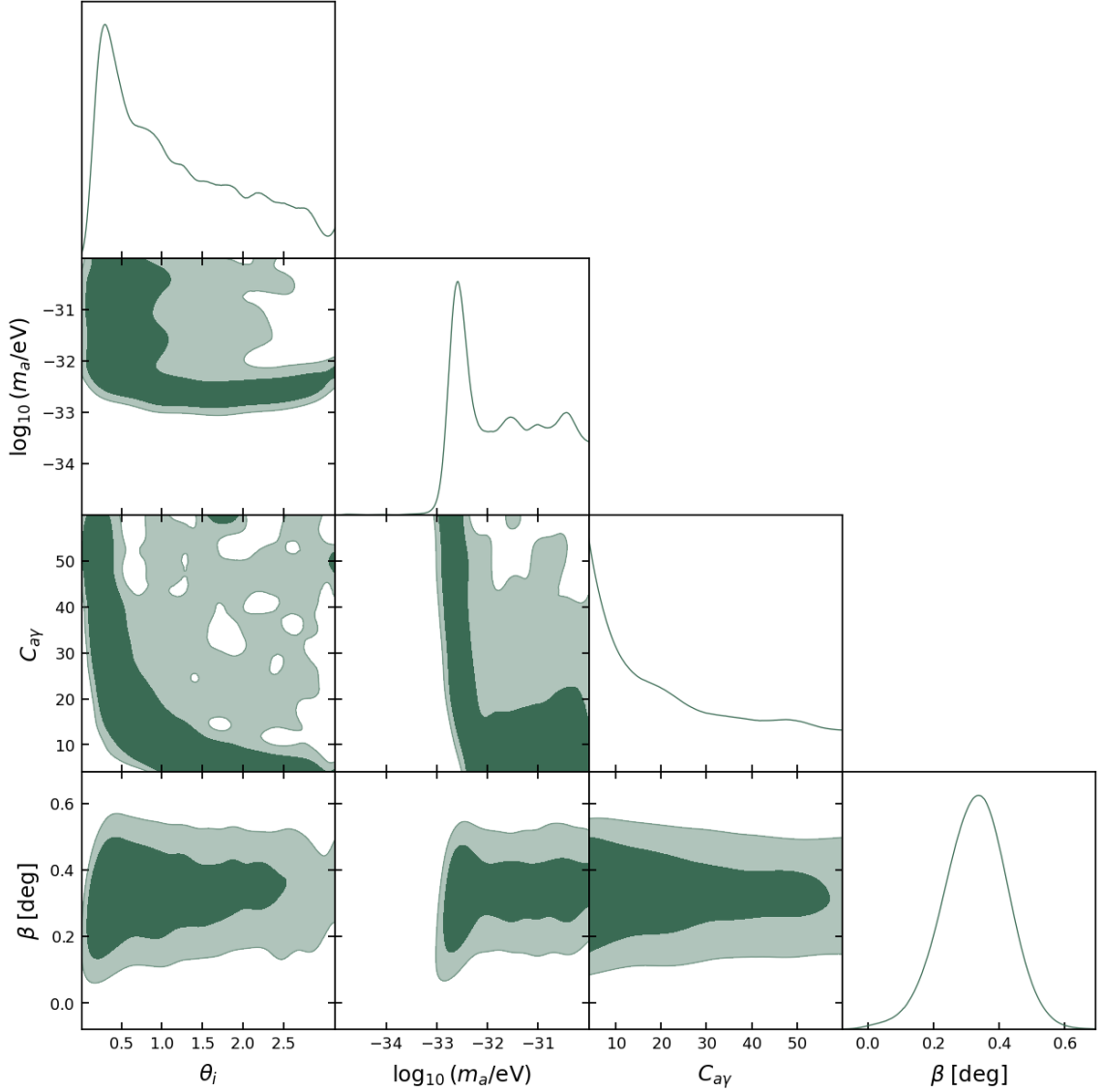


FIG. 4. Spectator-ALP joint posterior triangle from the continuous-prior cross-check configuration in which the photon anomaly coefficient is sampled freely (flat priors $C_{a\gamma} \in [4, 60]$ — shifted and extended from the earlier $[1, 30]$ to cover the posterior-supported coupling band (median 20.7, 16–84% $[7.3, 45.6]$); the dropped $[1, 4)$ interval lies entirely below the minimum coupling ≈ 8.6 required to reach the *central* value $\beta = 0.342^\circ$ at the largest in-box displacement $\Delta\phi/f_a = 1.19$ — couplings in $[4, 8.6)$ remain posterior-supported because the summary likelihood permits β below the central value — $\theta_i \in [0.01, \pi]$, $\log_{10}(m_a/\text{eV}) \in [-35, -30]$ — the mass prior corresponds to $m/H_0 \approx 7 \times 10^{-3}$ to 7×10^2 for $H_0 = 67.7 \text{ km/s/Mpc} = 1.44 \times 10^{-33} \text{ eV}$), broadening the fixed- $C_{a\gamma} = 8$ and $C_{a\gamma} \in [1, 30]$ configurations of Appendix C; the direct-sample priors are on the underlying ALP parameters (θ_i, m_a) , with $\Delta\phi/f_a$ derived along each trajectory. The rotation marginal, $\beta = 0.326^\circ \pm 0.099^\circ$, is consistent within 1σ with the fixed- $C_{a\gamma} = 8$ result $\beta_{\text{ALP}} = 0.336^\circ \pm 0.10^\circ$ and the observed $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$ (an internal-consistency statement: all three are constrained by the same single published β_{obs} , not independent measurements). The θ_i – $C_{a\gamma}$ anti-correlation band traces the constant-product degeneracy enforced by the summary-likelihood constraint through $C_{a\gamma}(\Delta\phi/f_a) \approx 10.3$, and the m_a marginal piles toward the upper (heavier) edge of its prior range — the data-driven pull toward the upper edge of the natural-prior box discussed in the text, consistent with the model *accommodating* rather than naturally predicting the observed rotation (fn. 6).

known limitation and no model-selection claim is made. The iter2 chain uses `planck_2018_lensing_native` in place of `planck_2018_lensing_clik` (Table III); a dedicated release-pairing swap test to assess the impact of this lensing likelihood substitution on the $w_0 w_a$ pos-

teriors has not been performed and is left to a post-submission follow-up.

Data and Code Availability

All materials are at:

<https://github.com/Hubify-Projects/bigbounce/tree/main/reproducibility>

The version of the repository corresponding to this paper is identified by the in-text v1B.0.81 stamp (defined in `paper1b.mcmc.companion.tex` `\newcommand{\paperVersion}`) and by the matching version-stamp commit in the repository git log (current snapshot commit: b22f8cc9); the per-version changelog (with the corresponding commit SHAs) is in the repository root `CHANGELOG.md`. The repository is program-wide: it includes the four Cobaya YAML configurations (one per dataset combination, stock CAMB, no custom modifications), NaMaster driver scripts, and the implementation map (`IMPLEMENTATION_MAP.md`), alongside galaxy-spin pipeline code used by Paper IV [8] but not by any analysis of this paper. Frozen MCMC chains are committed (HuggingFace datasets listed in Appendix A); fresh re-verification chains can be regenerated via `reproduce_cosmology.sh` (~4–12 h per configuration on 4 CPU cores). HuggingFace dataset URLs for the three accompanying datasets (MCMC chain diagnostics, NaMaster pipeline artifacts, ALP parameter chains) are listed below in Appendix A under “HuggingFace datasets”; DOI assignment is pending (identifiers will be inserted at submission). The URLs are also recorded in the repository `CHANGELOG.md` under the entry for v1B.0.81; links are preserved against future README changes via the version-stamp commit identified by the changelog. *Column-permutation warning for JSON artifacts.*—Readers should use `parameter\_summary\_CORRECTED.json` (the corrected diagnostic file in the frozen-chains directory) when extracting numerical values matching Table I; an earlier `parameter\_summary.json` export carried an off-by-one column-index bug. Full details of the bug, the corrected file, and the column-by-column mapping are documented in the repository `CHANGELOG.md` (v1B.0.59 archive entry) and in the adjacent `parameter\_summary\_units\_README.md`; this artifact-level process discussion is kept in the repository rather than in the paper proper.

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Appendix A: Reproducibility Materials

Repository structure.—The public repository <https://github.com/Hubify-Projects/bigbounce> contains:

- Four Cobaya YAML configurations (`cobaya_planck.yaml`, `cobaya_planck_bao.yaml`, `cobaya_planck_bao_sn.yaml`, `cobaya_full_tension.yaml`)—stock CAMB, no torsion modifications.
- `reproducibility/galaxy_spins/spin_fit_stan.py`—hierarchical Bayesian model (CmdStanPy) fitting $A(z)$ to published aggregate CW/CCW galaxy counts (program-wide content used by Paper IV, not by this paper’s analyses).
- `research/data_build/build_galaxy_spin_dataset.py`—reproducible pipeline downloading Galaxy Zoo DECaLS [29] from Zenodo (DOI: 10.5281/zenodo.4573248, CC-BY-4.0; likewise Paper IV content).
- `reproducibility/docs/IMPLEMENTATION_MAP.md`—mapping from each result to its code artifact.
- `reproducibility/docs/KNOWN_GAPS.md`—honest disclosure of what cannot currently be reproduced.

What is included vs. regenerable.—The two frozen Λ CDM+ ΔN_{eff} chain directories are committed in-repo and contain the chains and diagnostics that back Table I, Table III, and the relevant Table V claim-classification entries of this paper; the ALP c5 continuous chain backs Table IV separately: `reproducibility/cosmology/frozen/full_tension_20260311_1728/` and `reproducibility/cosmology/frozen/planck_bao_sn_20260312_1954/` (each with `chains/chain_0[1-6]/` and `diagnostics/parameter_summary_CORRECTED.json`). The converged iter2 $w_0 w_a$ chain bundle is committed at `reproducibility/cosmology/iter2_converged_2026-05-18/`, and the ALP-MCMC chains at `research/branch_R_alp_birefringence/phase2_mcmc/chains/`. Fresh Λ CDM+ ΔN_{eff} proxy chains for independent re-verification are NOT bundled and must be regenerated locally via `reproduce_cosmology.sh` (~4–12 h per config on 4 CPU cores). All Table I–II numerical entries quoted in this paper were recomputed directly from the committed raw chains (and from the `parameter_summary_CORRECTED.json` artifacts derived from them), not from the legacy column-permuted export. Bayes factors and information criteria (ΔAIC , ΔBIC , $\ln B$) are NOT reported in this manuscript: dedicated nested sampling via PolyChord or MultiNest on the identical likelihood stack is required for a

controlled Bayesian evidence on the present quinton posterior (Sec. V, *Model-comparison statistics* paragraph). A Savage-Dickey readout from the present Metropolis-Hastings chain is not viable—the Λ CDM point $(w, w_a) = (-1, 0)$ is unsampled (it lies deep in the joint marginal tails of the overlap-uncorrected posterior, but no σ -distance is quoted because the product-likelihood SN overlap narrows the widths), so a KDE-based estimator yields kernel-dependent noise rather than a controlled density. This is left to a dedicated future nested-sampling analysis. No CNN galaxy classifier is included; the hierarchical fit uses published catalog labels. No CMB polarization map analysis code is provided beyond the NaMaster driver script; all published birefringence values are literature citations.

HuggingFace datasets.—The following HuggingFace datasets accompany this work (DOI assignment is pending; identifiers will be inserted at submission. The URLs are also recorded in the repository `CHANGELOG.md` under the entry for `v1B.0.81` and are preserved against future README changes via the version-stamp commit identified by the changelog):

1. MCMC chain diagnostics and convergence CSV files: <https://huggingface.co/datasets/bamfai/p1b-mcmc-diagnostics>.
2. NaMaster pipeline artifacts (mask, MC seeds, output spectra): <https://huggingface.co/datasets/bamfai/p1b-namaster-artifacts>.
3. ALP parameter MCMC chains: <https://huggingface.co/datasets/bamfai/p1b-alp-chains>.

Appendix B: Claims Classification

Table V classifies the principal load-bearing quantitative claims made in this companion by claim type and verification status; it is the machine-checkable index used by the reproducibility audits of Appendix A.

Appendix C: ALP-MCMC Sampled Parameters, Priors, and Likelihood Stack

The ALP-MCMC results quoted in Sec. VI ($\beta_{\text{ALP}} = 0.336^\circ \pm 0.10^\circ$ at $C_{a\gamma} = 8$ fixed; $\beta_{\text{free}} = 0.344^\circ \pm 0.10^\circ$ model-independent; 9,720 total accepted samples across the three committed configurations) use the following setup. All priors below are read directly from the archived chain configurations (`research/branch_R_alp_birefringence/phase2_mcmc/chains/`); the headline posteriors are verified directly against the committed chains.

Configuration (i) — fixed-coupling fit (run1_full; 2,160 accepted samples).

- $C_{a\gamma}$: fixed at 8 (theory-module default; not sampled).
- θ_i : uniform prior on $[0.01, \pi]$.⁷
- $\log_{10}(m_a/\text{eV})$: uniform prior on $[-35, -30]$ ($m/H_0 \approx 7 \times 10^{-3}$ to 7×10^2).
- f_a : fixed at M_{Pl} (spectator-class theoretical input from the Holst-sector pseudoscalar structure; not sampled).

This configuration yields $\beta_{\text{ALP}} = 0.336^\circ \pm 0.10^\circ$ with posterior $\theta_i = 1.32 \pm 0.41$ and median $m \approx 36 H_0$.

Configuration (ii) — sampled-coupling fit (run2_extended; 6,840 accepted samples).—As configuration (i) with $C_{a\gamma}$ additionally sampled, uniform prior $[1, 30]$. The $[4, 60]$ continuous-prior rerun (below) is the primary coupling-inference result; the $[1, 30]$ prior truncated $\sim 28\%$ of the coupling posterior mass.

Configuration (iii) — model-independent β_{free} fit (run3_baseline; 720 accepted samples).

- β : uniform prior on $[-1^\circ, 2^\circ]$; sampled as a free amplitude with no ALP-model parametric structure. Yields $\beta_{\text{free}} = 0.344^\circ \pm 0.10^\circ$.

Sampled parameters and priors (continuous-prior cross-check configuration; Fig. 4).—Three sampled parameters (read directly from the archived chain configuration, `research/branch_R_alp_birefringence/phase2_mcmc/chains/c5_continuous/c5.input.yaml`):

- $C_{a\gamma}$: uniform prior on $[4, 60]$ (covering the posterior-supported band $[9, 51]$ that contains 69% of the posterior mass, see Sec. VI; the full kinematic natural-box requirement extends to $C_{a\gamma} \approx 160$ at the smallest misalignment displacements, as discussed in the same section).
- θ_i : uniform prior on $[0.01, \pi]$.
- $\log_{10}(m_a/\text{eV})$: uniform prior on $[-35, -30]$ ($m/H_0 \approx 7 \times 10^{-3}$ to 7×10^2).

f_a is not a sampled parameter in this chain. The likelihood is the same Eskilt–Komatsu Gaussian summary as above; 8,955 accepted samples, $\hat{R} - 1 = 0.0095$.

Effective sample sizes (ESS).—Per-parameter ESS computed from the integrated autocorrelation time of the weight-expanded chains (Sokal estimator; `tools/ess_alp_chains.py`):

⁷ Backreaction / spectator-status disclosure: see fn. 6 in Sec. VI. The wide prior is an envelope-completeness choice, not the spectator-consistent sub-range ($\theta_i \sim 0.1$, a $\sim 25\times$ tuning relative to the natural midpoint); posterior samples at $\theta_i \gtrsim 0.5$ belong to the dark-energy-ALP regime excluded from the spectator-consistency claim.

TABLE V. Claims classification for this companion paper. The “Reference value” column identifies the public source or internal verification anchor for each claim; “Int. verified” means the quoted value has been reproduced from committed chains/artifacts at the present **v1B.0.81** commit (**b22f8cc9**); a public tagged release is pending (see Data and Code Availability).

Claim	Type	Reference value	Notes
$\Delta N_{\text{eff}} = -0.020 \pm 0.169$ (full-tension)	MCMC	Int. verified (frozen chains)	Stock CAMB proxy
$\Delta N_{\text{eff}} = +0.058 \pm 0.179$ (Planck+BAO+SN)	MCMC	Int. verified (frozen chains)	Stock CAMB proxy
$H_0 = 67.68 \pm 1.06$ (full-tension)	MCMC	Int. verified; Planck-consistent	Recovers Λ CDM
$H_0 = 67.78 \pm 1.09$ (Planck+BAO+SN)	MCMC	Int. verified; Planck-consistent	Recovers Λ CDM
Model-comparison $\Delta\text{AIC/BIC}/\ln B$	Numerical	Not reported	Nested-sampling follow-up
$\hat{\beta}_{\text{NaMaster}} = 0.238^\circ$ (500-MC)	Numerical	Int. verified (pipeline)	Obs. pipeline bias
$\beta_{\text{ALP}} = 0.336^\circ \pm 0.10^\circ$	MCMC	Int. verified (ALP chains)	ALP MCMC
Published 3.6σ ($\beta = 0.342 \pm 0.094^\circ$)	Lit.	[5]	Eskilt et al.
Stock CAMB proxy $\neq$ ECH theory module	Scope	Definition	§III
ALP birefringence not distinctive ECH prediction	Scope	Definition	§VI

Parameter	Chain	N_{acc}	ESS
θ_i	c5_continuous	8,955	796
$\log_{10}(m_a/\text{eV})$	c5_continuous	8,955	1,114
$C_{a\gamma}$	c5_continuous	8,955	814
β_{deg} (derived)	c5_continuous	8,955	2,860
β_{free}	run3_baseline	720	265

The **c5_continuous** ESS values (~ 800 – 2860) are adequate for posterior characterisation. The β_{free} **run3_baseline** chain (720 accepted samples, $\text{ESS} \approx 265$) is marginal; the result $\beta_{\text{free}} = 0.344^\circ \pm 0.10^\circ$ should be interpreted with the caveat that ~ 720 accepted samples provides limited ESS for this single-parameter fit.

Likelihood stack.—All fits use a Gaussian summary likelihood on the published Eskilt–Komatsu joint WMAP+Planck isotropic-birefringence measurement $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$ [5] (fn. 3; the chain configuration encodes exactly **beta_obs: 0.342, sigma.beta: 0.094**) — i.e. the constraint enters at the level of the published β posterior, not as a direct re-analysis of the *EB* spectra. The separate ACT DR6 measurement $\beta = 0.215^\circ \pm 0.074^\circ$ [4] is an independent cross-check quoted in Sec. VI and does *not* enter any MCMC

likelihood. The MCMC engine is Cobaya v3.6.1 with the Metropolis-Hastings sampler; convergence threshold $\hat{R} - 1 < 0.01$ across configurations (i)–(iii) ($N_{\text{tot}} = 9,720$ accepted samples) and for the continuous-prior $C_{a\gamma} \in [4, 60]$ configuration ($\hat{R} - 1 = 0.0095$, 8,955 accepted samples). *Chain-to-number map:* $\beta_{\text{ALP}} = 0.336^\circ \pm 0.10^\circ$ — the fixed- $C_{a\gamma} = 8$ chain **run1_full** (2,160 samples); $\beta_{\text{free}} = 0.344^\circ \pm 0.10^\circ$ — the model-independent single-parameter chain **run3_baseline** (720 samples); median $C_{a\gamma} = 20.7$, 16–84% range [7.3, 45.6], $\beta = 0.326^\circ \pm 0.099^\circ$, and the spectator-subset readouts — the continuous-prior **c5** chain (8,955 samples); the $C_{a\gamma} \in [1, 30]$ chain **run2_extended** (6,840 samples) is retained for the prior-truncation comparison only.

Scope statement.—These fits constrain a specific ultra-light ALP class ($m \sim H_0$, $f_a \sim M_{\text{Pl}}$; the $C_{a\gamma} \in [4, 12]$ benchmark sweep plus the continuous-prior $C_{a\gamma} \in [4, 60]$ headline configuration). The consistency with β_{obs} established here is *not* a universal mechanism-independent statement; alternative parity-violating mechanisms (Chern-Simons, axion dark matter at higher m , etc.) require independent fits with their own model parameters.

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